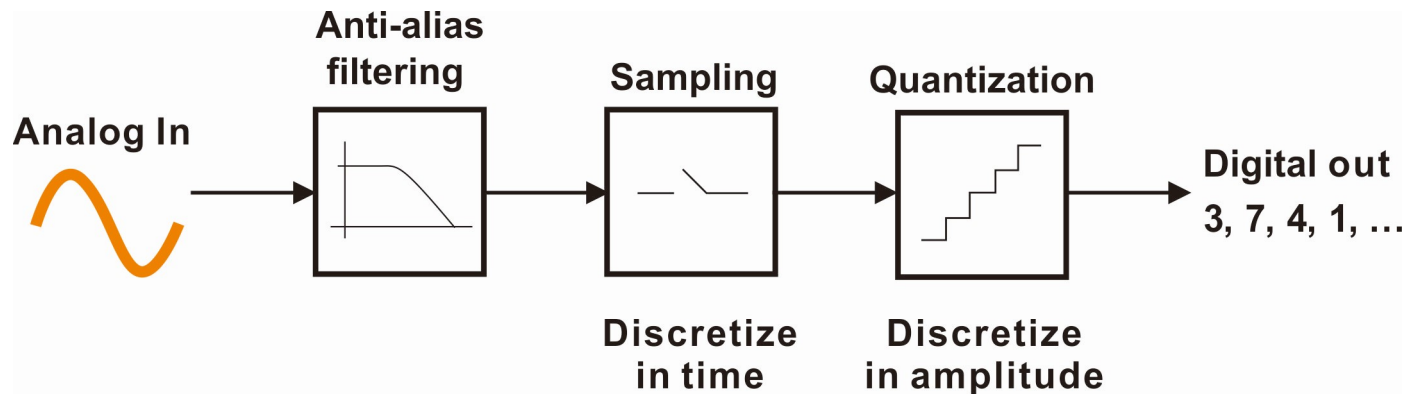


Sampling Circuits

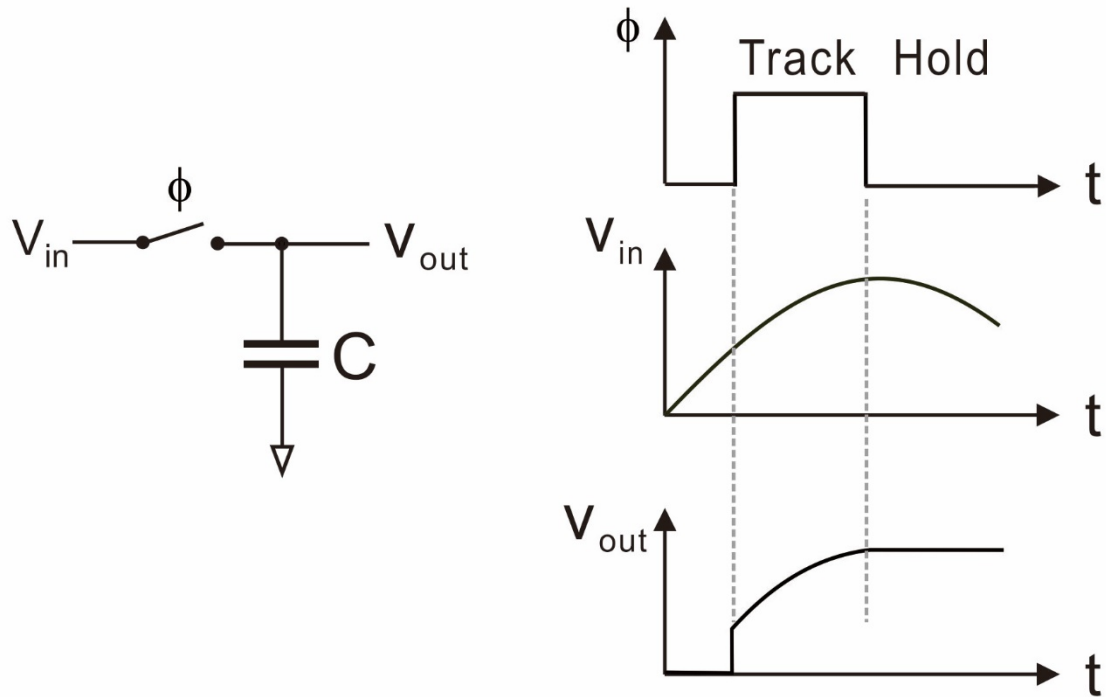
Nan Sun
Tsinghua University
nansun@tsinghua.edu.cn

ADC Big Picture



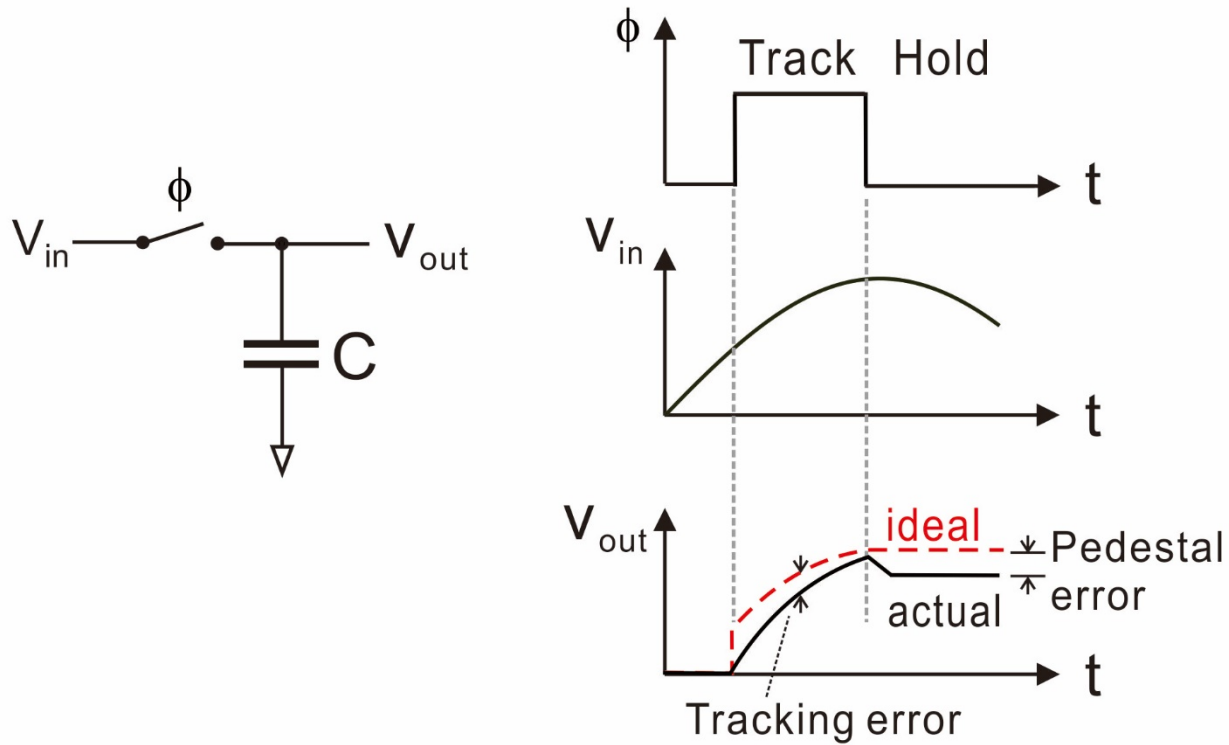
- How to build a sampling circuit in the real-world?
 - Ideal Dirac sampling is impractical
 - Build a track and hold circuit

Ideal Track-and-Hold Circuit



- Infinitely fast and accurate tracking of V_{in}
- No error in the hold phase

Circuit with MOS Switch



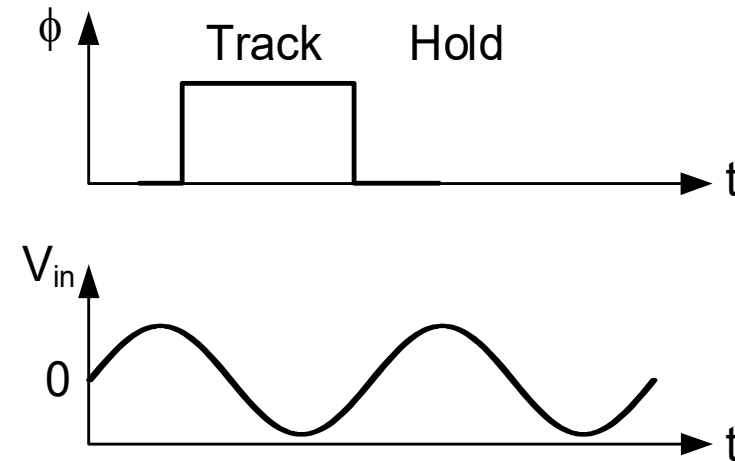
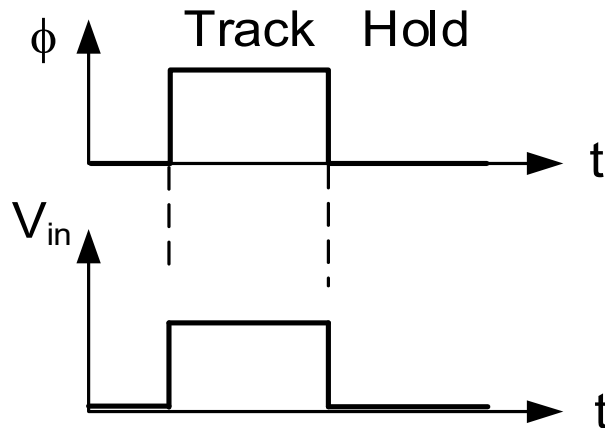
- Tracking error
- Pedestal error

Nonidealities

- Switch resistance
 - Finite acquisition time
 - Thermal noise
 - Nonlinear resistance
- Switch capacitance
- Sampling clock jitter
- Other nonidealities

Finite Acquisition Time

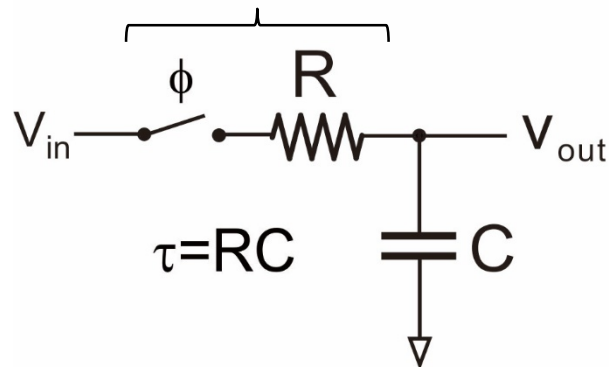
- Case 1: Input is a sampled data signal, e.g. the output of a switched capacitor circuit
- Case 2: Input is a varying continuous-time signal



Finite Acquisition Time – Case 1

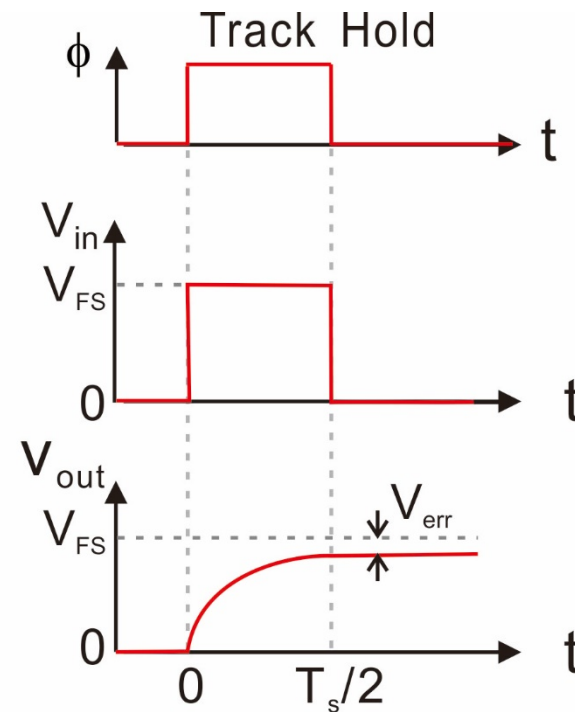
- For simplicity, let us ignore the finite rise time of V_{in}
- Consider the worst case: V_{out} settles from 0 to V_{FS}

1st-order MOS switch model



$$V_{out}(t) = V_{FS}(1 - e^{-t/\tau})$$

$$V_{err} = V_{out}(T_s/2) - V_{FS} = -V_{FS}e^{-T_s/2\tau}$$



Finite Acquisition Time – Case 1

- We want the settling error $|V_{\text{err}}| < \frac{1}{2} \text{ LSB}$?

$$V_{\text{FS}} e^{-T_s/2\tau} \leq \Delta/2 = V_{\text{FS}}/2^{B+1}$$

- The required # time constant N:

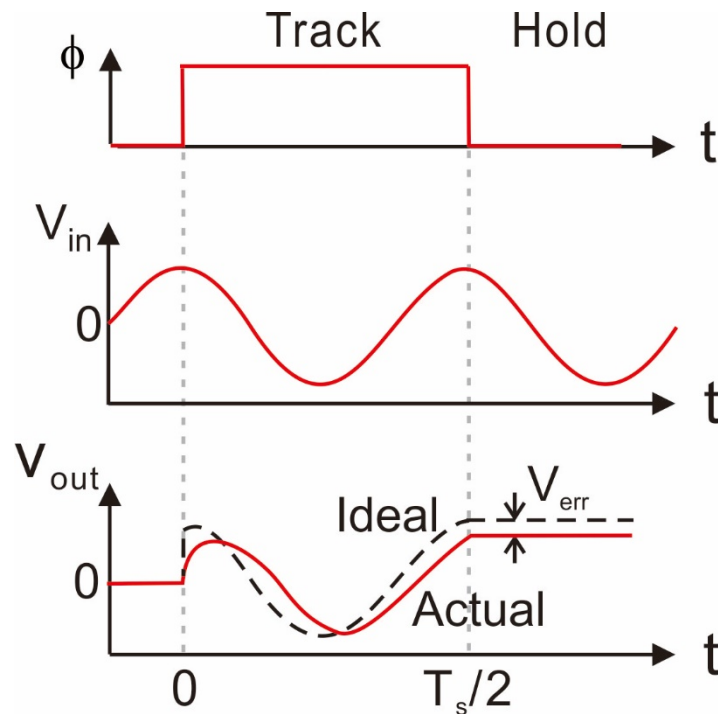
$$N \equiv \frac{T_s}{2\tau} > \ln(2^{B+1})$$

- Does settling error cause distortion?

B	N
6	>4.9
10	>7.6
14	>10.4
18	>13.2

Finite Acquisition Time – Case 2

- Consider a sinusoidal input with zero offset



$$V_{in}(t) = A \cos(\omega t)$$

$$V_{out}(t) \approx -Ae^{-t/\tau} + \frac{A}{\sqrt{1 + \omega^2 \tau^2}} \cos(\omega(t - \tau))$$

Initial
discontinuity
settling

Steady-state
RC filter response

Finite Acquisition Time – Case 2

- The error at $t=T_s/2$ consists of two parts:
 - Error due to the exponential decay of the initial discontinuity
 - Similar to Case 1, it decays with time
 - Error in the steady-state term
 - Result of RC LPF
 - Magnitude attenuation and phase shift
 - It depends on RC (filter response) and the input frequency
 - It does not decay with time
 - Will this error cause distortion?
 - How large is the amplitude attenuation?

Finite Acquisition Time – Case 2

- Let us calculate the percent amplitude error

$$A_{\text{err}} = 1 - \frac{1}{\sqrt{1 + (\omega\tau)^2}} = 1 - \frac{1}{\sqrt{1 + \left(\frac{\pi f_{\text{in}}}{N f_s}\right)^2}} \approx \frac{1}{2} \left(\frac{\pi f_{\text{in}}}{N f_s}\right)^2$$

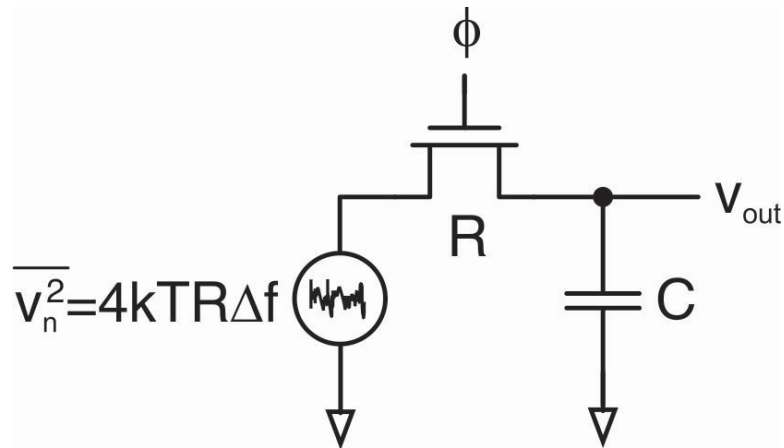
- Amplitude error increases with f_{in} , but decreases with N .

B	N	$A_{\text{err}} (f = f_s/10)$	$A_{\text{err}} (f_{\text{in}} = f_s/2)$
6	4.9	0.20%	5%
10	7.6	0.08%	2%
14	10.4	0.05%	1%
18	13.2	0.03%	0.7%

Summary – Finite Acquisition Time

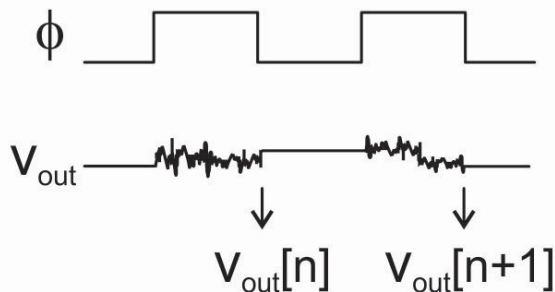
- For sampled-data input, we can achieve accurate tracking given a sufficient amount of time.
- For continuous input, we need to consider the errors due to both the initial settling and **the filter response**.
 - In applications where attenuation is intolerable, we may need to further reduce the RC time constant.
 - In applications where attenuation is tolerable, the RC time constant is usually set by the distortion specs.
 - **The larger the attenuation, the larger the instantaneous voltage drop across the nonlinear MOSFET** → undesired harmonics.
- What about the phase delay?
 - The sampled input is always a delayed (τ) version.

Thermal Noise



- Transistor thermal noise dominates
- $1/f$ noise can be ignored
 - $1/f$ noise is like an offset

$$\frac{\overline{v_{out}^2}}{\Delta f} = 4kTR \left| \frac{1}{1 + sRC} \right|^2$$



$$\overline{|v_{out}^2[n]|} = 4kTR \int_0^{\infty} \frac{1}{1 + (2\pi fRC)^2} df = \frac{kT}{C}$$

Equipartition Theorem

- Equipartition theorem in statistical physics states that in thermal equilibrium each degree of freedom holds on average thermal energy of $kT/2$.
- In an electronic circuit, each independent capacitor holds an electric energy of $kT/2$ and each independent inductor holds a magnetic energy of $kT/2$.
- This theorem can help reduce the complexity of noise analysis.
- Use this theorem cautiously!

$$\begin{aligned}\frac{1}{2} C \overline{v_{out}^2} &= \frac{1}{2} kT \\ \Rightarrow \overline{v_{out}^2} &= \frac{kT}{C}\end{aligned}$$

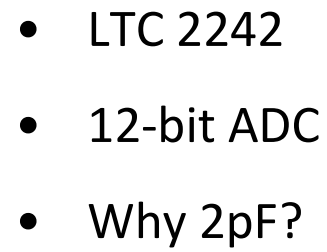
kT/C Noise

- Assuming that we make the kT/C noise equal to the quantization noise of a B-bit ADC

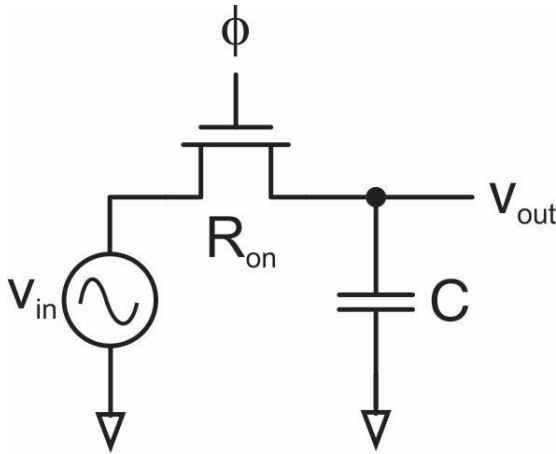
$$\frac{kT}{C} = \frac{\Delta^2}{12} \quad \Rightarrow \quad C = 12kT \cdot \left(\frac{2^B}{V_{FS}} \right)^2$$

- Assuming $V_{FS} = 1V$,

B	C [pF]
8	0.003
10	0.052
12	0.834
14	13.3
16	213
18	3416



Voltage Dependence of R_{on}



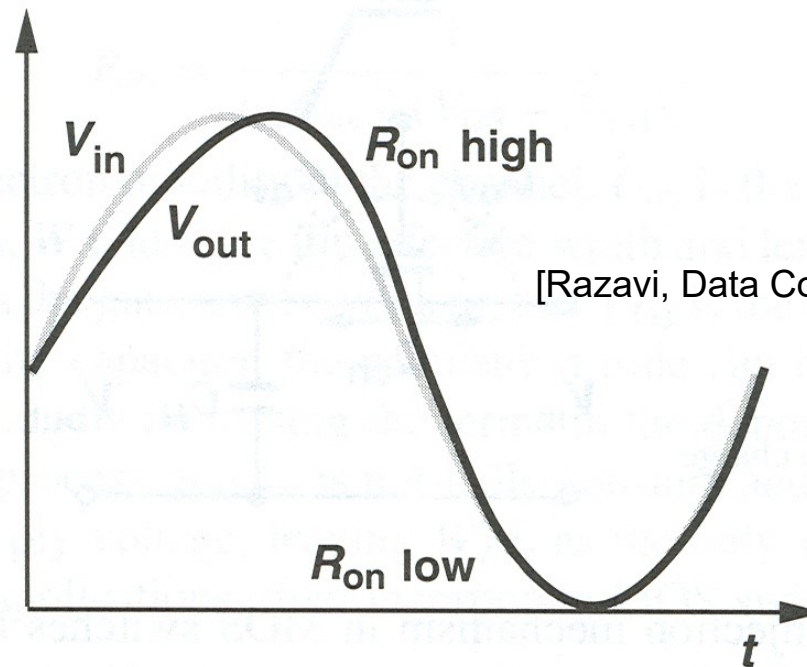
$$I_D = \mu C_{ox} \frac{W}{L} \left(V_{GS} - V_{th} - \frac{V_{DS}}{2} \right) V_{DS}$$

$$R_{ON} \approx \left[\frac{\partial I_D}{\partial V_{DS}} \Big|_{V_{DS} \rightarrow 0} \right]^{-1} = \frac{1}{\mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})}$$

$$R_{ON} \approx \frac{1}{\mu C_{ox} \frac{W}{L} (\phi - V_{IN} - V_{th})}$$

- R_{ON} is dependent on V_{in}

Track Mode Nonlinearity



[Razavi, Data Conversion System Design, p.16]

- V_{out} tracks V_{in} well for small V_{in}
 - Large $V_{in} \rightarrow$ large $R_{on} \rightarrow$ distortion

Analysis

$$I_D \approx K(V_{GS} - V_{th})V_{DS} - \frac{K}{2}V_{DS}^2$$

$$C \frac{dV_{out}}{dt} = I_D \approx K(\varphi - V_{out} - V_{th})(V_{in} - V_{out}) - \frac{K}{2}(V_{in} - V_{out})^2$$

- We can solve the above equation use Volterra Series analysis
 - General method that allows us to calculate the frequency domain response of nonlinear circuits with memory
 - See [Yu, TCAS II, 2/1999]

Result

$$\text{HD}_3 = \frac{\text{Amplitude of the third harmonic}}{\text{Amplitude of the fundamental}} \cong \frac{1}{4} \frac{A^2}{(V_{\text{GS}} - V_{\text{th}})^2} \cdot \omega\tau$$

- V_{GS} is the value of the gate-source voltage when AC input is zero.
- For low distortion
 - Make amplitude A smaller than $(V_{\text{GS}} - V_{\text{th}})$
 - Low swing $\rightarrow$ SNR decreases
 - Make ω smaller
 - Low frequency $\rightarrow$ speed decreases
 - Make $\tau = RC$ smaller
 - Big switch $\rightarrow$ parasitic caps increases and driving power goes up

Example

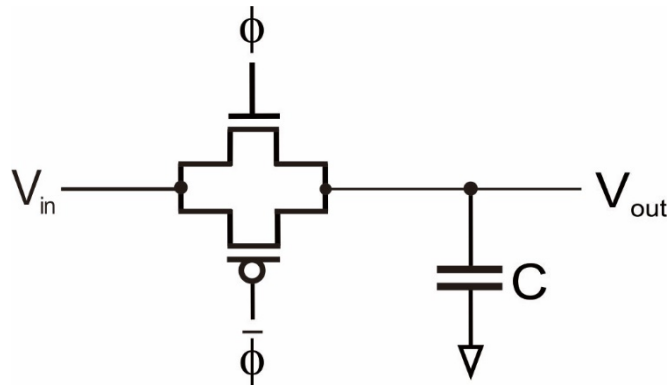
- Parameters
 - $V_{DD} = V_{CLK} = 1.8V$
 - V_{in} is centered at $V_{DD}/2 = 0.9V$
 - $V_{GS} - V_{th} = 1.8V - 0.9V - 0.45V = 0.45V$
 - $A = 0.4V$
 - $N = 0.5T_s/\tau = 10$
 - $f_{in} = f_s/2$

$$HD_3 \approx \frac{\pi}{4} \frac{0.4^2}{(0.45)^2} \cdot \left(\frac{1}{20}\right) = -30 \text{ dB}$$

- The nonlinearity is huge. How can we reduce it?

CMOS Switch

- Adding a PMOS can reduce the resistance variation
 - As V_{IN} increases, NMOS resistance goes up, but PMOS resistance goes down



$$R \approx \frac{1}{\mu_n C_{ox} \left[\frac{W}{L} \right]_n (V_{GSn} - V_{thn})} \parallel \frac{1}{\mu_p C_{ox} \left[\frac{W}{L} \right]_p (|V_{GSp}| - |V_{thp}|)}$$

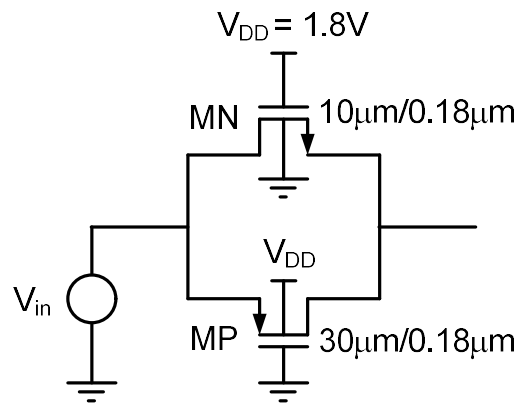
Analysis

$$R \approx \frac{1}{\mu_n C_{ox} \left[\frac{W}{L} \right]_n (V_{DD} - V_{thn}) - \left(\mu_n C_{ox} \left[\frac{W}{L} \right]_n - \mu_p C_{ox} \left[\frac{W}{L} \right]_p \right) V_{in} - \mu_p C_{ox} \left[\frac{W}{L} \right]_p |V_{thp}|}$$
$$\approx \frac{1}{\mu_n C_{ox} \left[\frac{W}{L} \right]_n (V_{DD} - V_{thn} - |V_{thp}|)}$$

if $\mu_n \left[\frac{W}{L} \right]_n = \mu_p \left[\frac{W}{L} \right]_p$

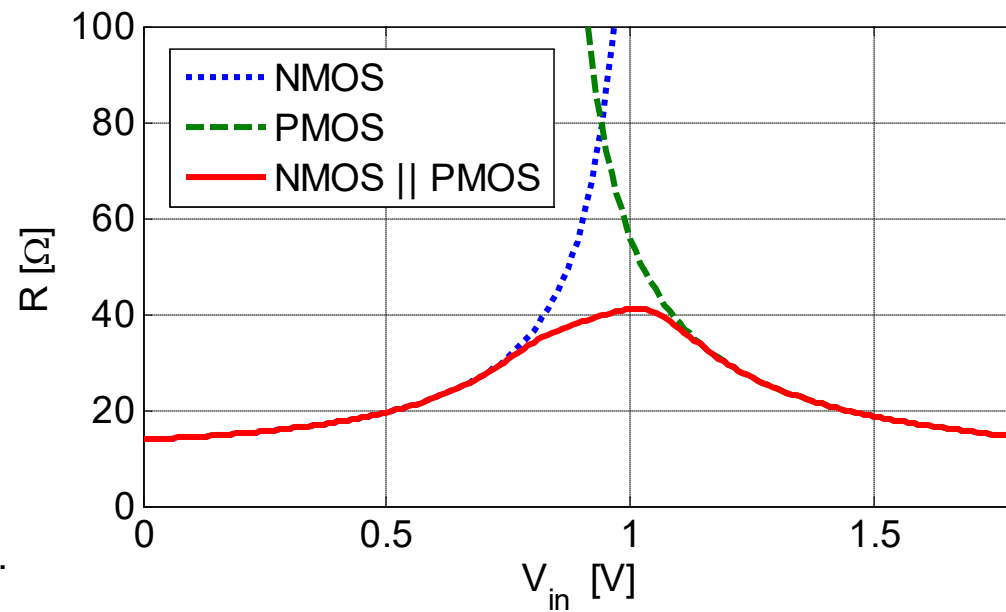
- R is independent from V_{in} ; however, this model is over-simplified.
- Missing factors
 - Short channel effects
 - Back-gate effect

Real CMOS Switch



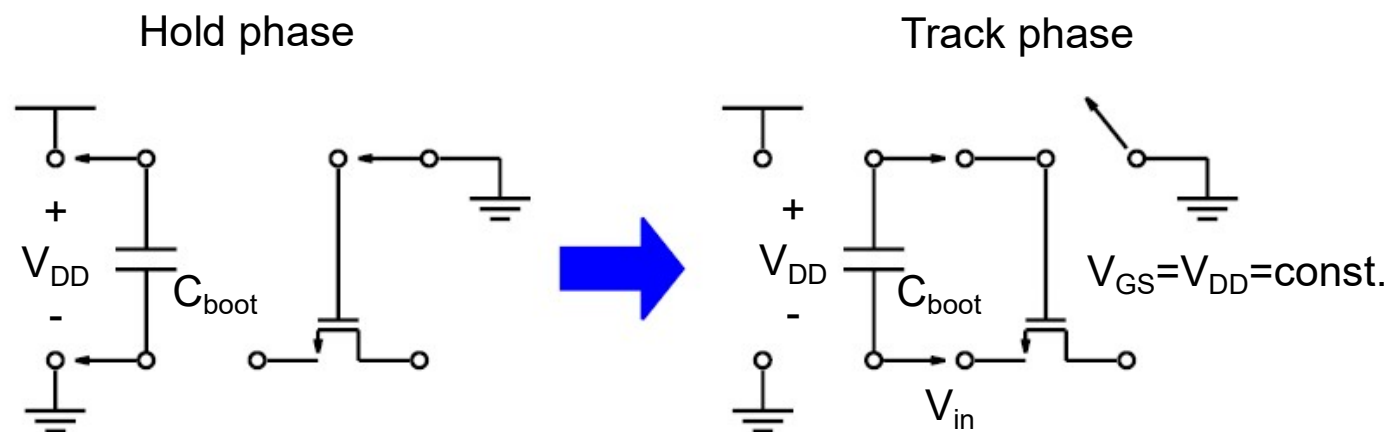
From B. Murmann, EE 315, Ch.

5



- There are still large resistance variations. The benefit is limited.

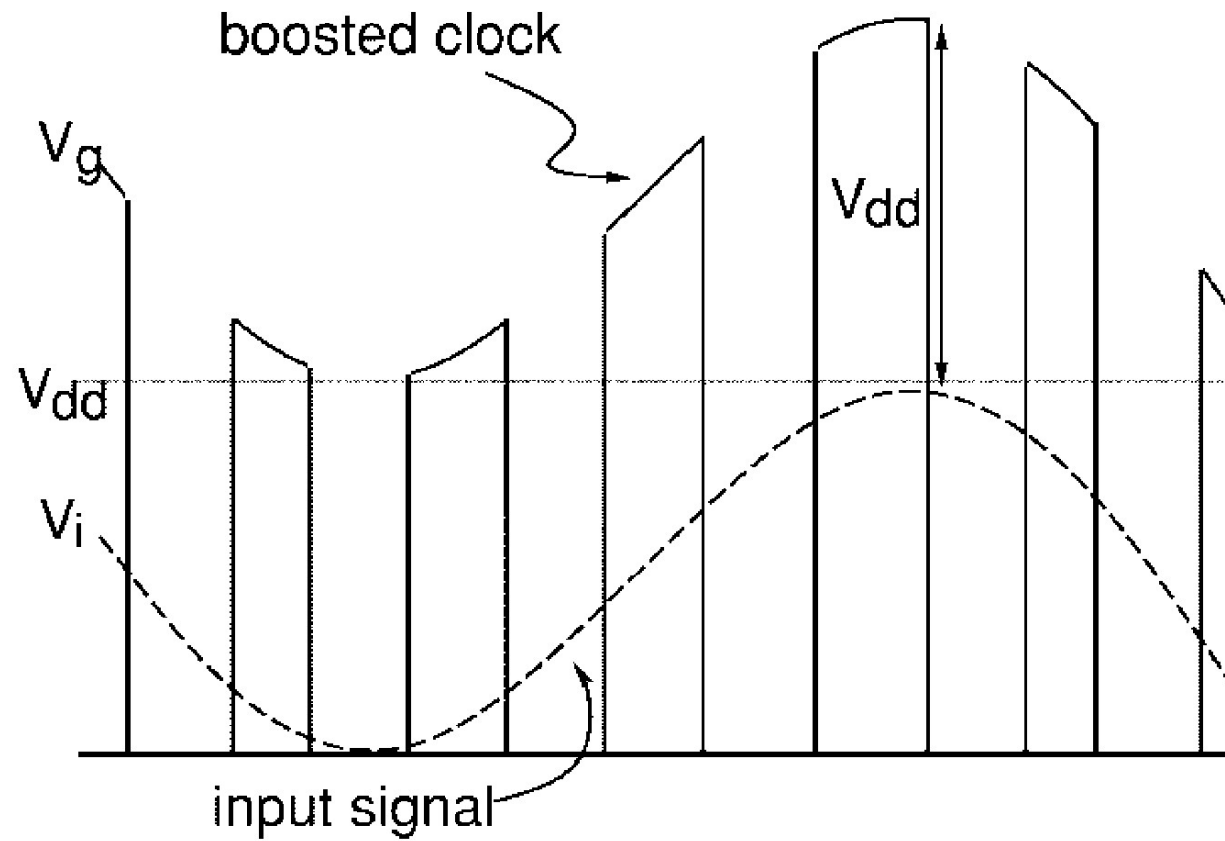
Clock Bootstrapping



A. Abo, "Design for Reliability of Low-voltage, Switched-capacitor Circuits," PhD Thesis, UC Berkeley, 1999.

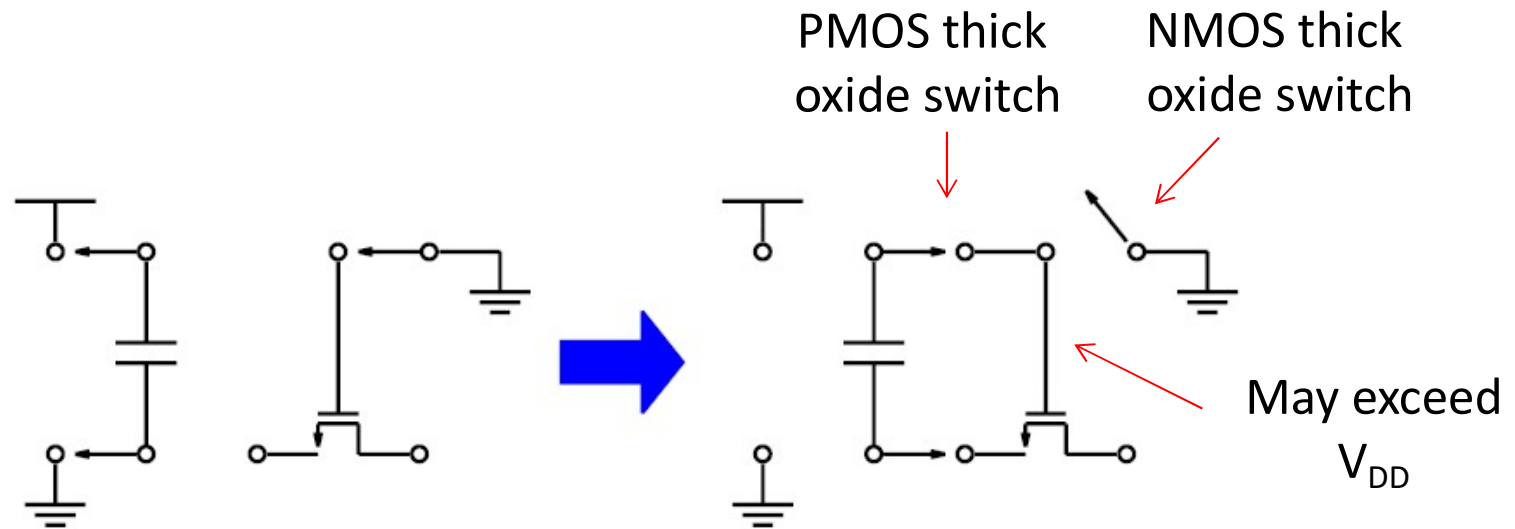
- Hold phase
 - Sampling switch is off
 - C_{boot} is precharged to V_{DD}
- Track phase
 - Sampling switch is on with $V_{GS} = V_{DD} = \text{const}$
 - Thus, R_{on} is signal independent

Waveforms



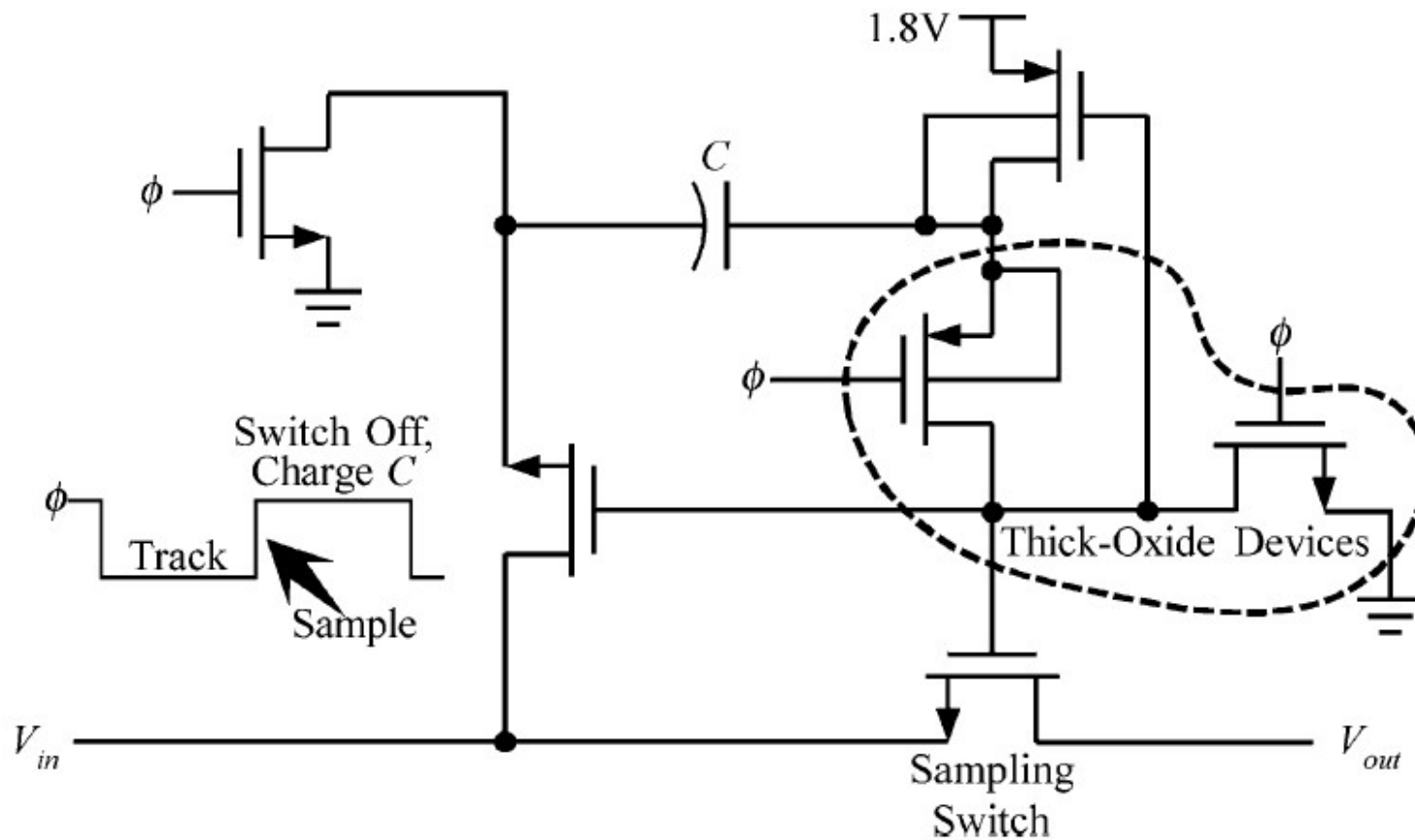
A. Abo, "Design for Reliability of Low-voltage, Switched-capacitor Circuits," PhD Thesis, UC Berkeley, 1999.

Implementation Issues



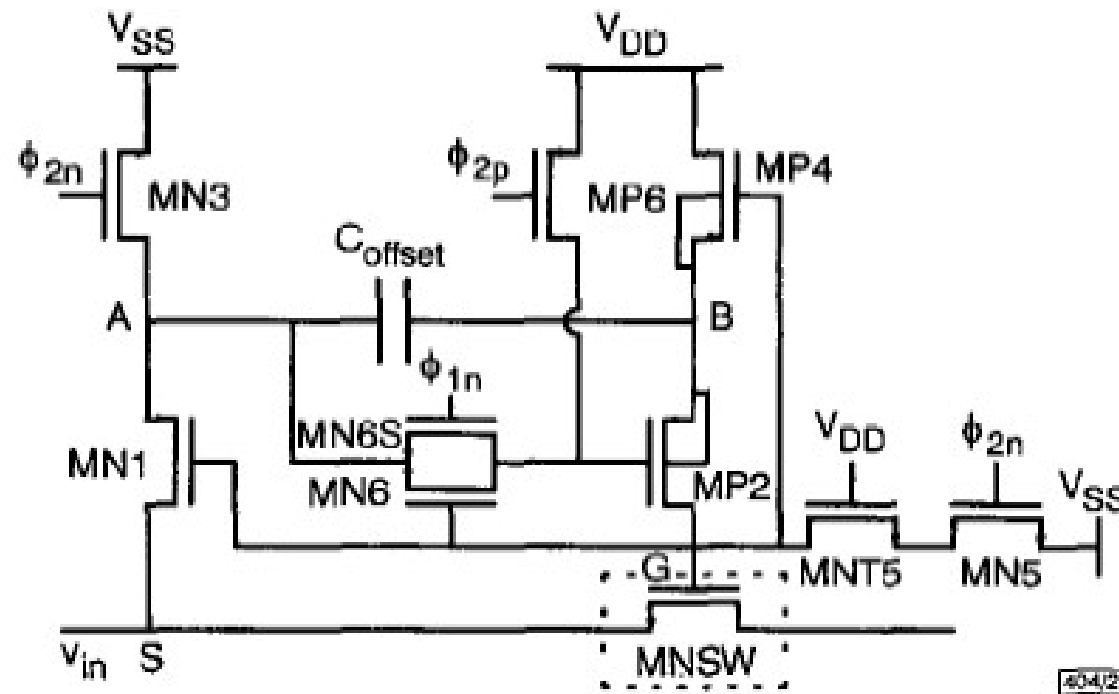
- V_{GS} or V_{GD} may exceed V_{DD} , causing transistor breakdown.
- Thick oxide devices can be used to ensure reliability, leading to increased cost, larger parasitic capacitance, and larger resistance.

With Thick-Oxide Devices

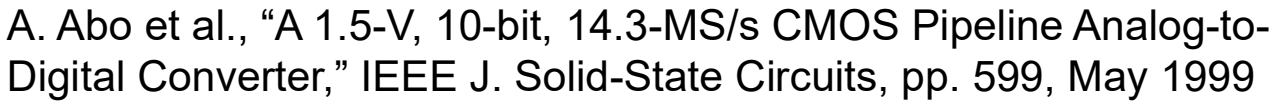


Siragusa and Galton, "A digitally enhanced 1.8-V 15-bit 40-Msample/s CMOS pipelined ADC," JSSC, Dec. 2004.

Without Thick-Oxide Devices

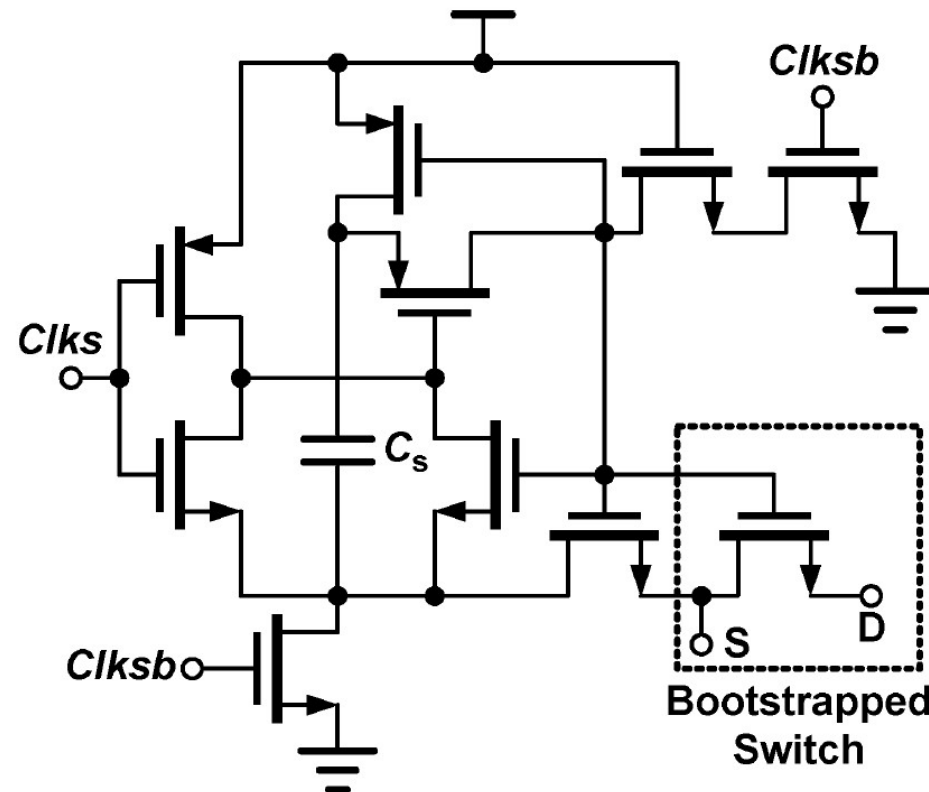


Dessouky and Kaiser, "Input switch configuration suitable for rail-to-rail operation of switched opamp circuits," Electronics Letters, Jan. 1999.



Simplified Implementation

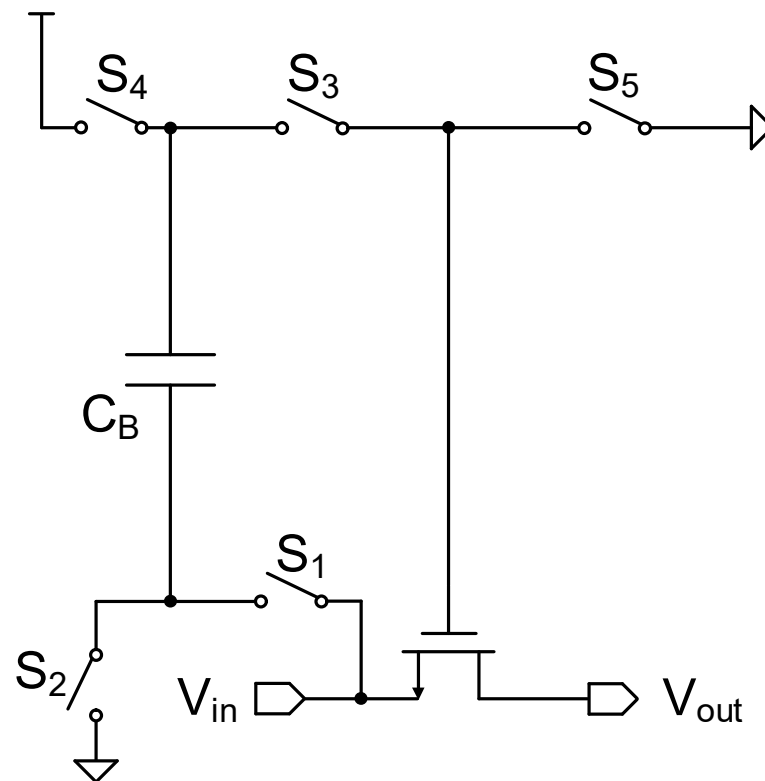
- An incremental design procedure to complete the schematic
 - B. Razavi, "The Design of a Bootstrapped Sampling Circuit [The Analog Mind]"



C.-C. Liu, "A 10-bit 50-MS/s SAR ADC with a monotonic capacitor switching procedure," IEEE J. Solid-State Circuits, pp. 731, 2010

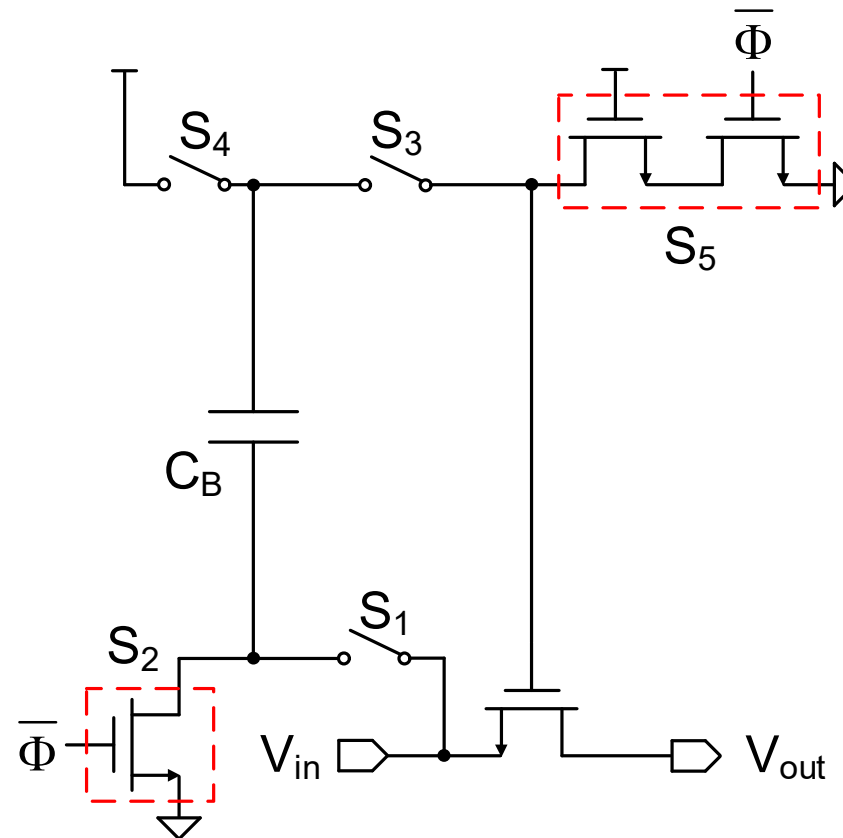
Design Procedure

- A complete bootstrapped switch model
 - 1 capacitor with 5 switches
 - Track mode: S_1 and S_3 on
 - Hold mode: S_2 , S_4 , and S_5 on



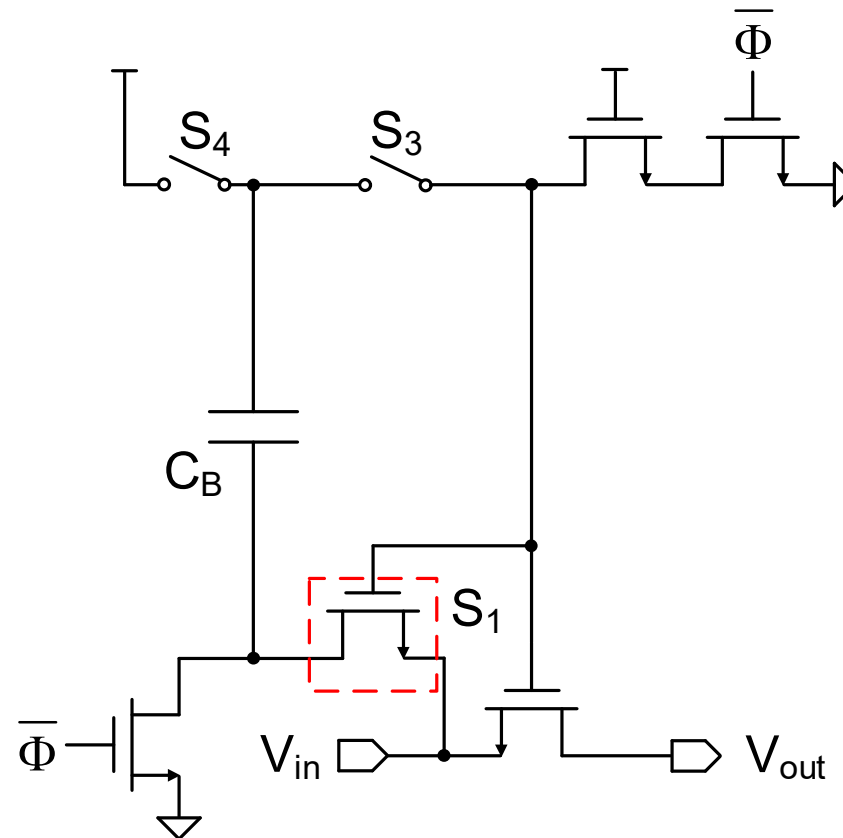
Design Procedure

- S_2 and S_5 must be NMOS devices because they pull nodes to ground
 - Cascode implementation of S_5 avoids using thick-oxide device



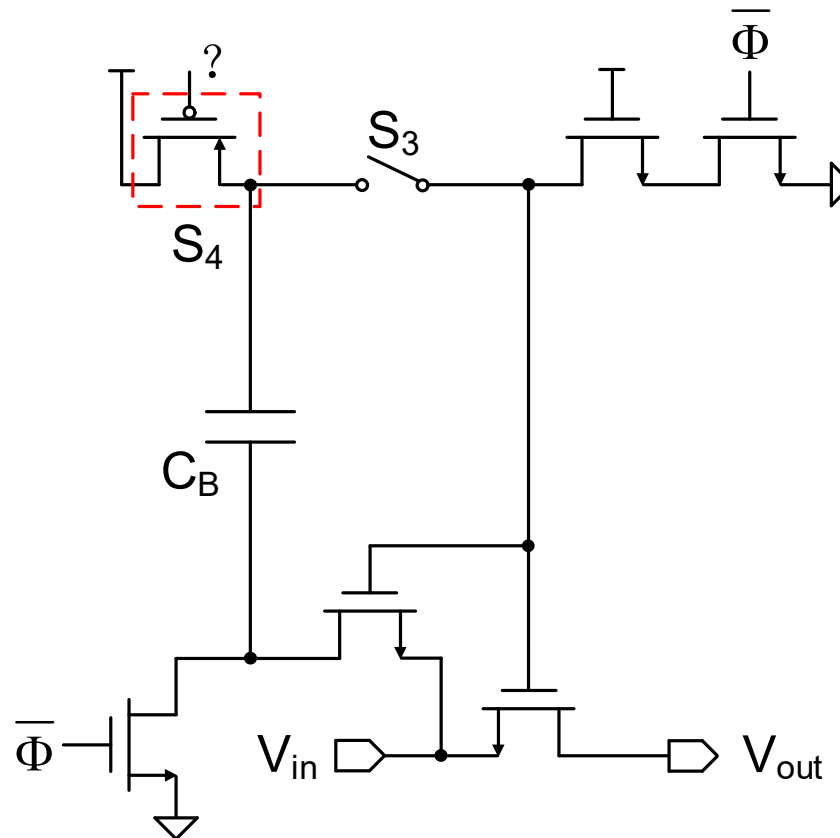
Design Procedure

- The gate of S_1 should be bootstrapped similar to the main switch
 - If connected to Φ ($0 \sim V_{DD}$), S_1 will turn off with large input swing



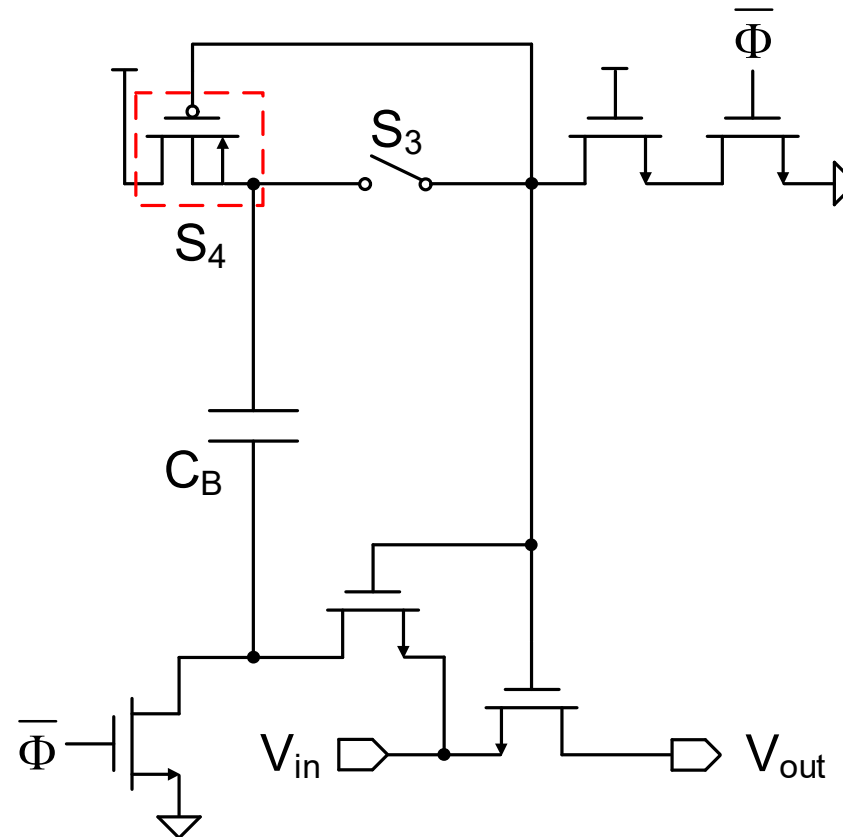
Design Procedure

- S_4 must be a PMOS device to charge C_B in the hold mode
 - Body connection?
 - Gate connection?



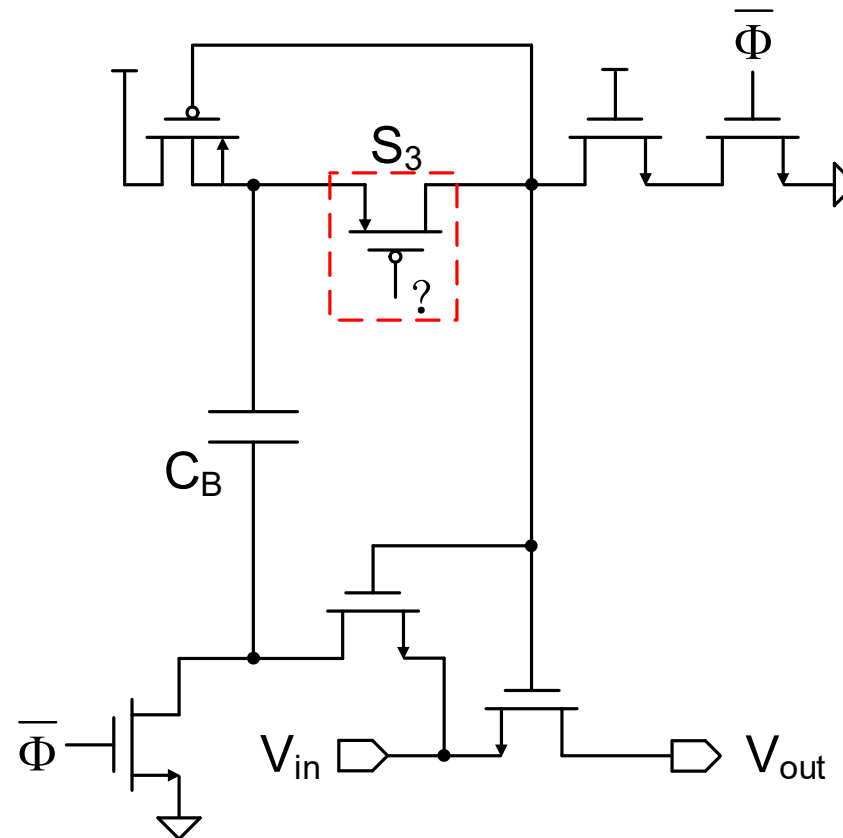
Design Procedure

- S_4 must be a PMOS device to charge C_B in the hold mode
 - The body should be connected to terminal with the highest voltage
 - The gate should be bootstrapped
 - If connected to Φ ($0 \sim V_{DD}$), S_4 fails to turn off in the track mode



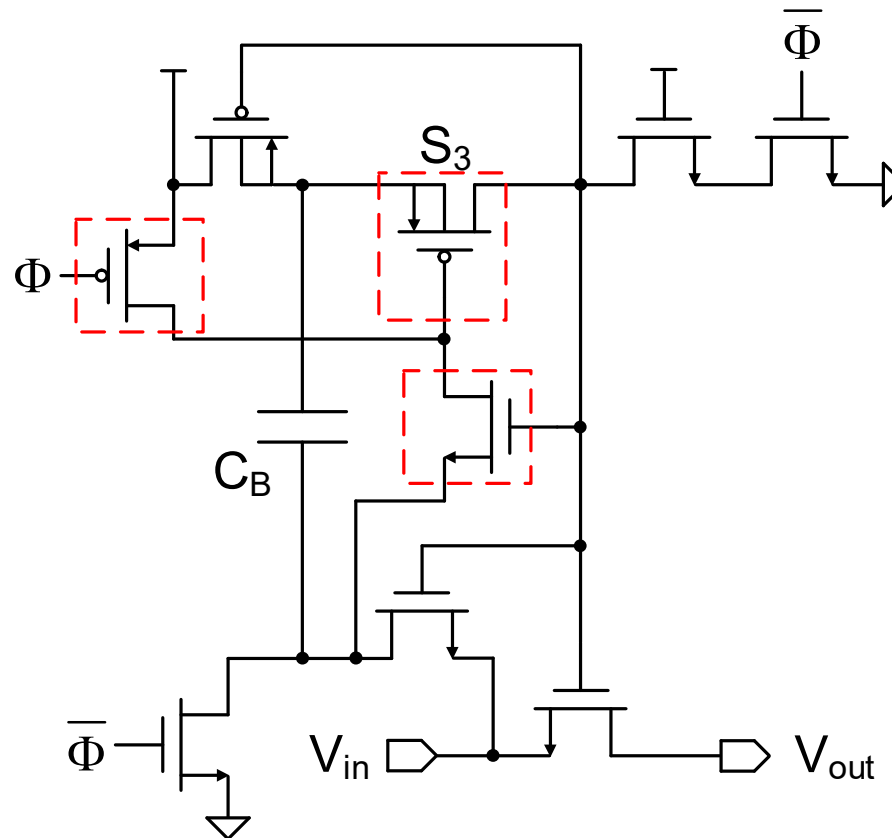
Design Procedure

- S_3 should be a PMOS device to pull up the gate of the main switch
 - Hold mode: $V_{\text{gate}} = V_{\text{DD}}$
 - Track mode: $V_{\text{gate}} = V_{\text{in}}$ to avoid using thick-oxide device
 - Body connection?



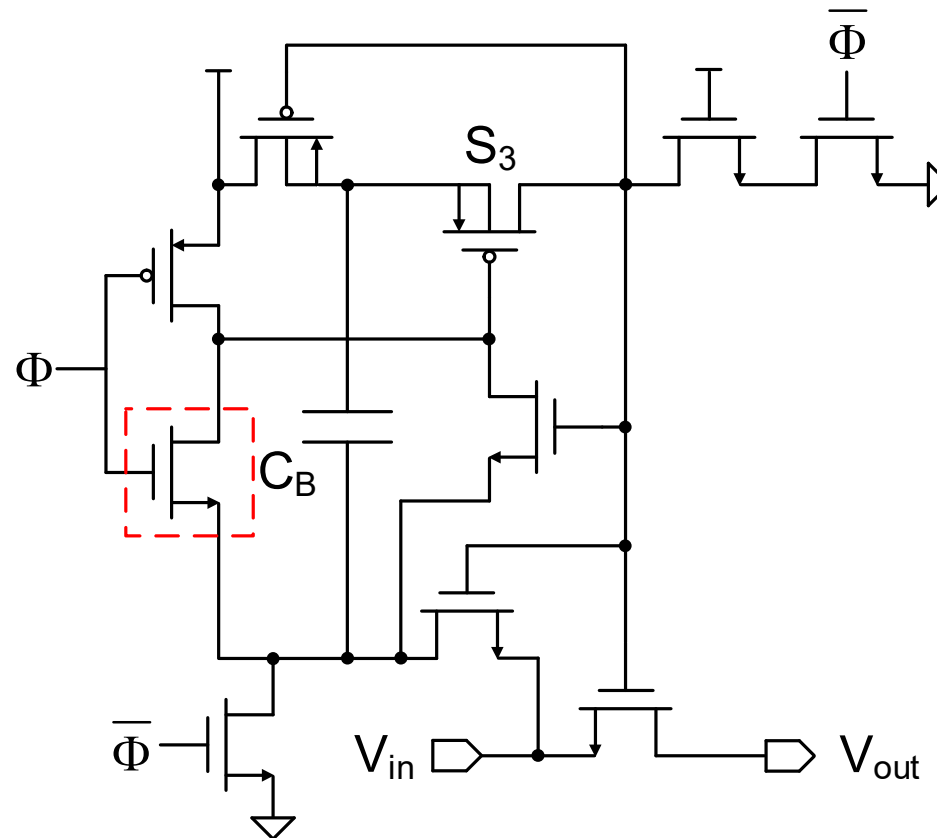
Design Procedure

- S_3 should be a PMOS device to pull up the gate of the main switch
 - Hold mode: $V_{\text{gate}} = V_{\text{DD}}$
 - Track mode: $V_{\text{gate}} = V_{\text{in}}$ to avoid using thick-oxide device
 - Body should be connected to terminal with the highest voltage



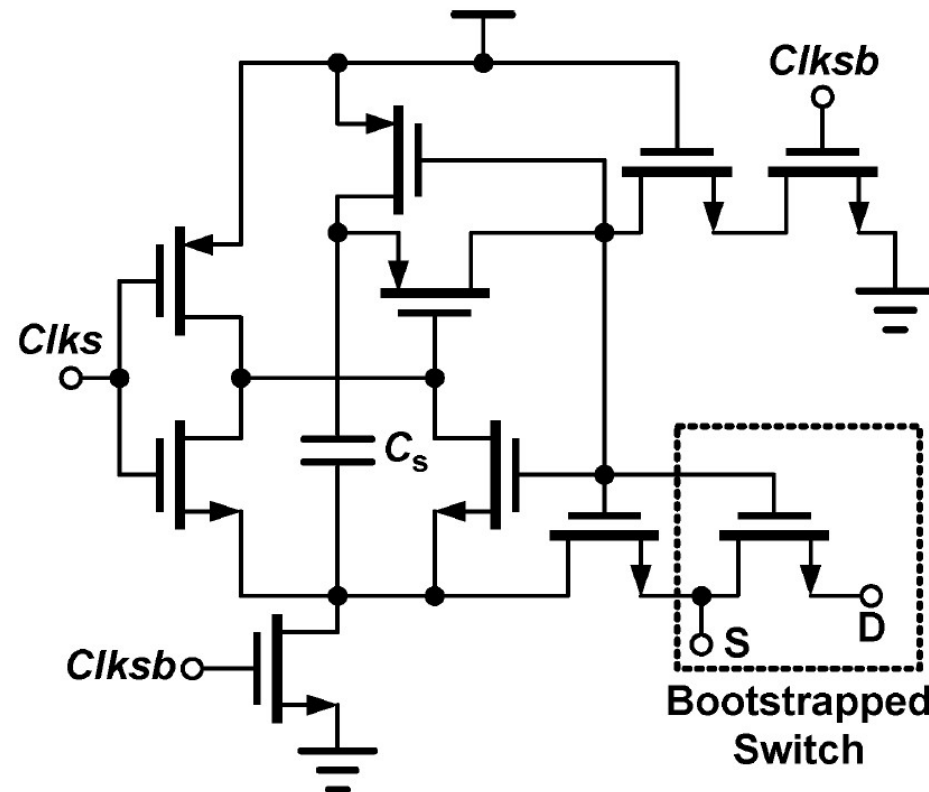
Design Procedure

- An extra NMOS device is necessary for the start-up of the bootstrapped circuit
 - Initially pull down the gate of S_3



Design Procedure

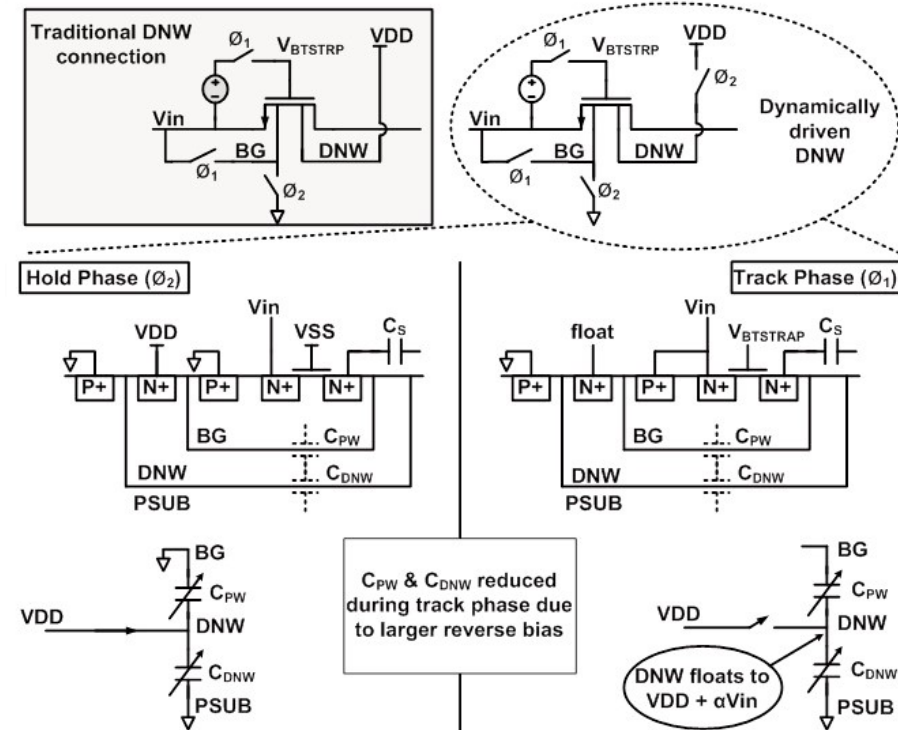
- Finally, here is what we've got!



C.-C. Liu, "A 10-bit 50-MS/s SAR ADC with a monotonic capacitor switching procedure," IEEE J. Solid-State Circuits, pp. 731, 2010

Limitation of Bootstrap Technique

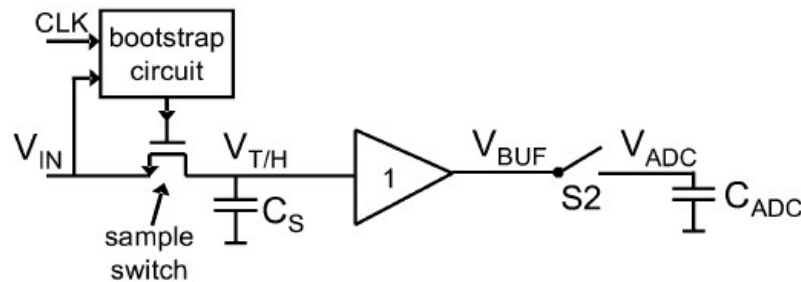
- Parasitic capacitance at the gate of the sampling switch
 - Capacitive divider $\rightarrow V_{GS}$ not a constant
 - Can be alleviated by increasing the size of the bootstrap capacitor
 - However, limited by the intrinsic parasitic cap associated with the bootstrap cap
- Back-gate effect
 - Can be alleviated by using triple well process



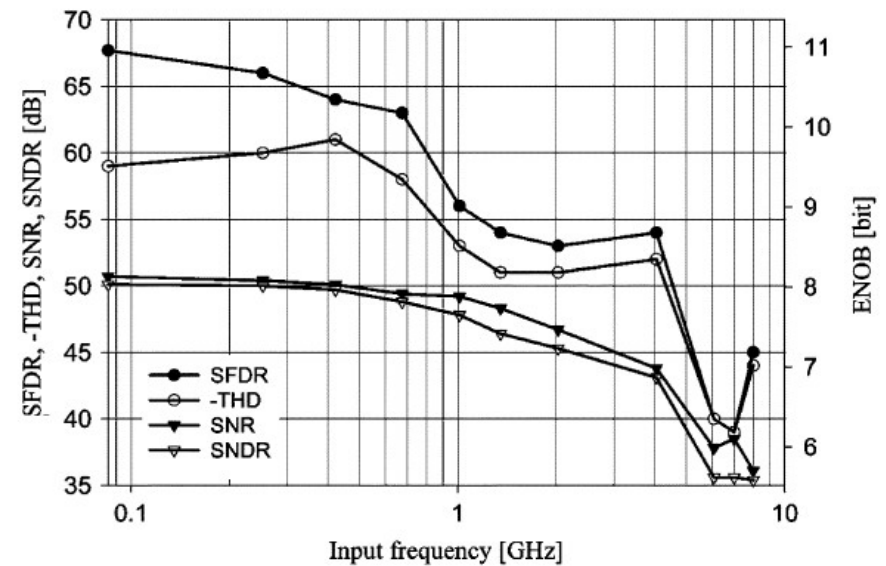
Brunsilus, ISSCC 2011, 10.4, pp. 186

Performance of Bootstrapped Sampler

- Bootstrapped sampling tends to work well up to 60 dB linearity
 - Now the linearity bottleneck is related to the switch capacitance.
- Example



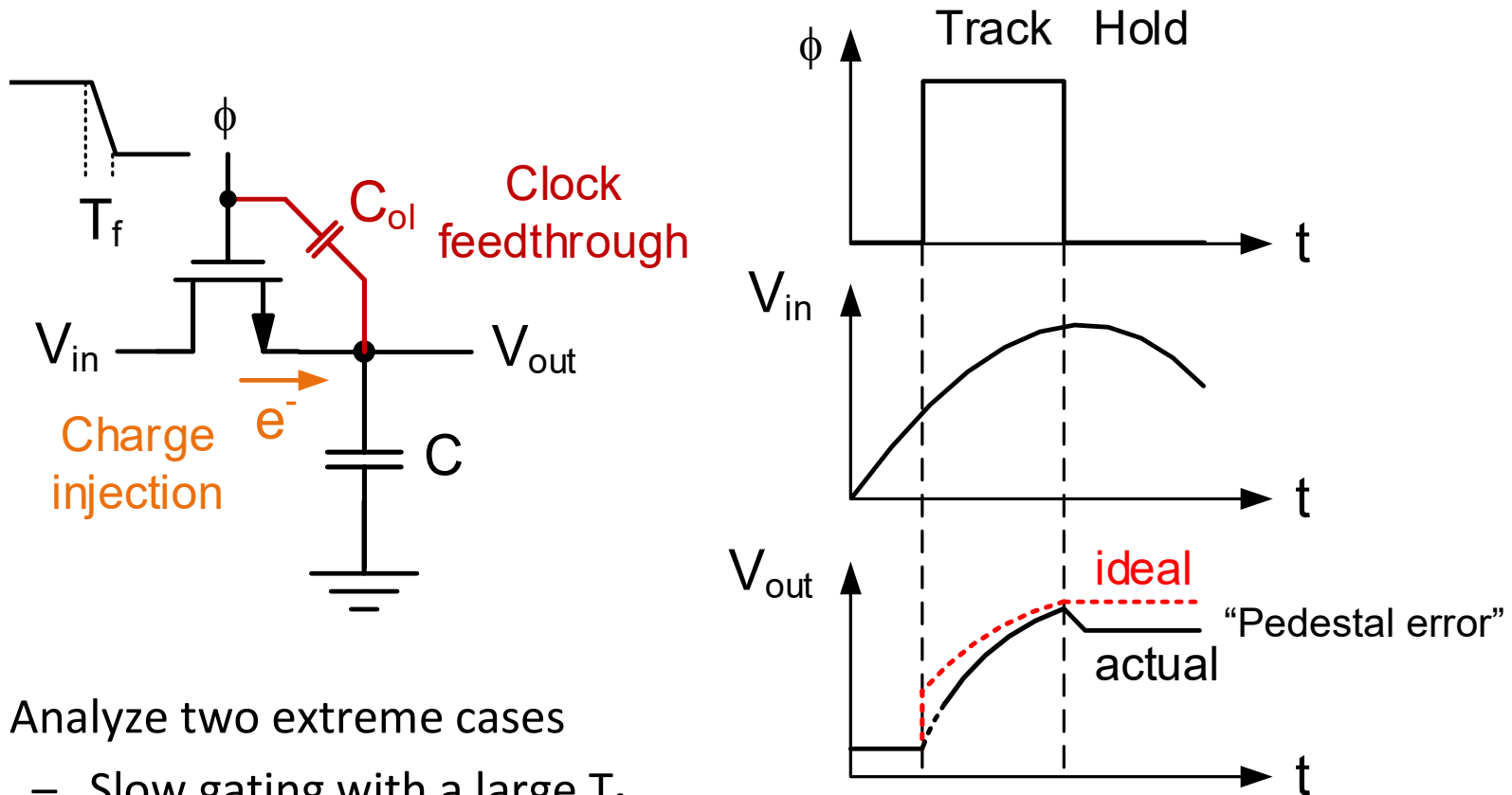
[Louwsma, JSSC 4/2008]



Nonidealities

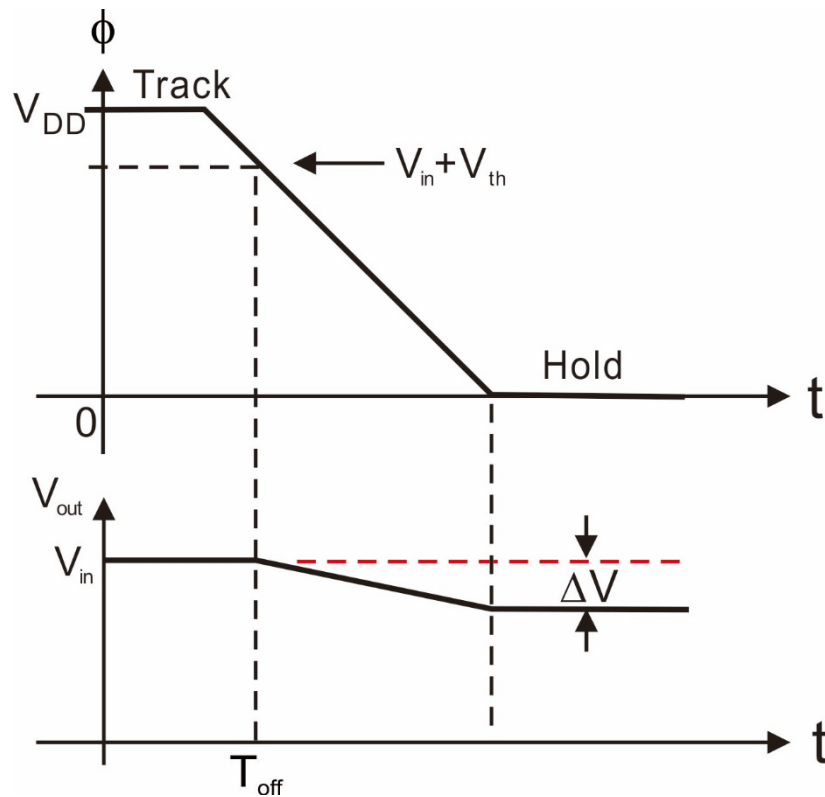
- Switch resistance
- Switch capacitance
 - Clock feedthrough
 - Channel charge injection
- Sampling clock jitter
- Other nonidealities

Charge Injection & Clock Feedthrough



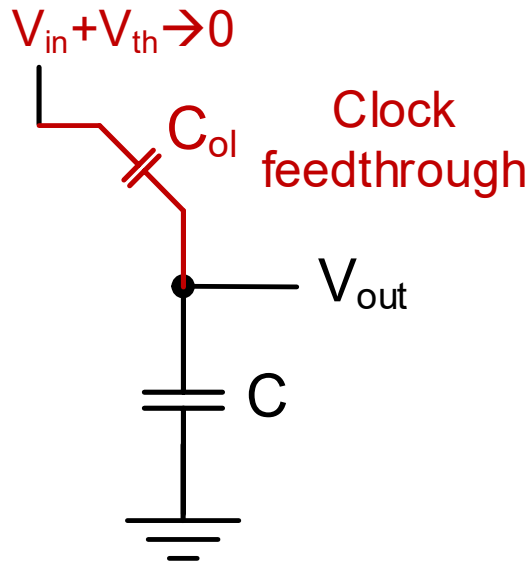
- Analyze two extreme cases
 - Slow gating with a large T_f
 - Fast gating with a small T_f

Slow Gating



- All channel charges have disappeared by T_f . They are absorbed by the input source, and thus, do not introduce error.
- Pedestal error is only due to clock feedthrough.

Slow Gating Model for $t > T_f$



$$V_{out} = V_{in} - \Delta V = V_{in} - \frac{C_{ol}}{C_{ol} + C} (V_{in} + V_{th})$$

$$= V_{in}(1 + \varepsilon) + V_{os}$$

switch open

$$\varepsilon = -\frac{C_{ol}}{C_{ol} + C}$$

Gain Error

$$V_{os} = -\frac{C_{ol}}{C_{ol} + C} V_{th}$$

Offset Error

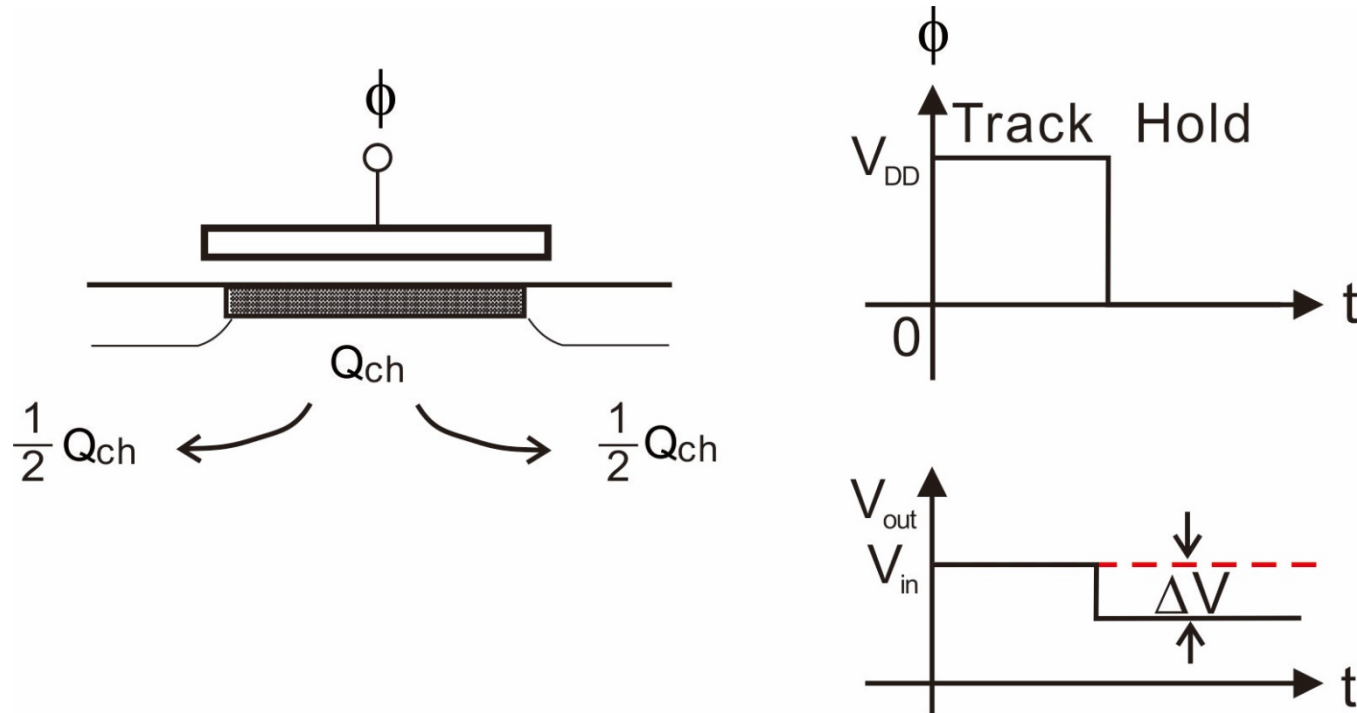
- Example: $C=1\text{pF}$, $C_{ol}=2\text{fF}$, and $V_{th}=0.45\text{V}$

$$\varepsilon = -0.2\%$$

$$V_{os} \approx -1\text{mV}$$

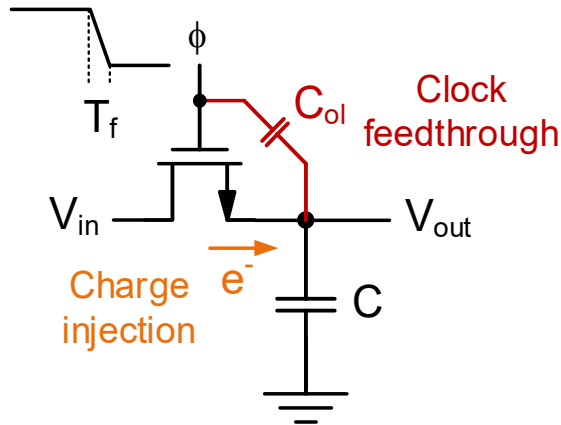
- Does this cause nonlinearity?

Fast Gating



- Channel charge Q_{ch} has to flow to the input and the output nodes
- Under fast gating, we assume a 50/50 split

Fast Gating Model for $t > T_f$



$$V_{out} = V_{in} - \frac{C_{ol}}{C_{ol} + C} V_{DD} + \frac{Q_{ch}}{2C} = V_{in}(1 + \varepsilon) + V_{os}$$

$$\text{where } Q_{ch} = -C_{ox}WL(V_{DD} - V_{in} - V_{th})$$

$$\varepsilon = \frac{1}{2} \frac{C_{ox}WL}{C}$$

$$V_{os} = -\frac{C_{ol}}{C_{ol} + C} V_{DD} - \frac{1}{2} \frac{C_{ox}WL}{C} (V_{DD} - V_{th})$$

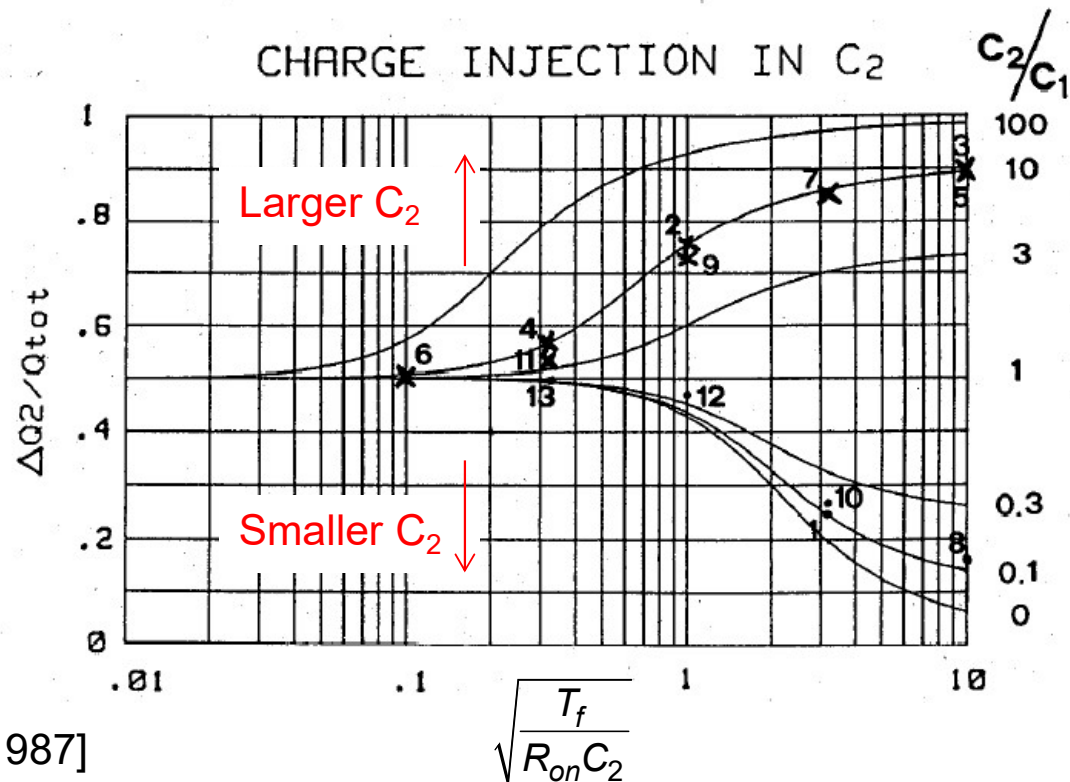
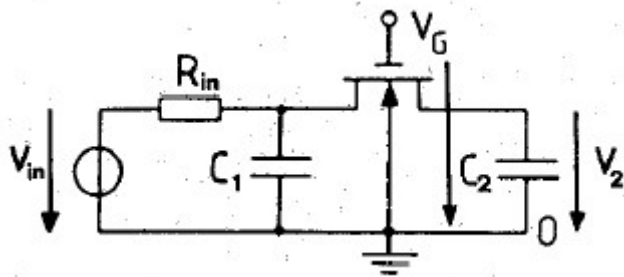
- Example: $C=1\text{pF}$, $V_{DD}=1.8\text{V}$, $V_{th}=0.45\text{V}$, $W=20\mu\text{m}$, $LC_{ox}=2\text{fF}/\mu\text{m}$, and $C_{ol}=2\text{fF}$

$$\varepsilon = 2\%$$

$$V_{os} \approx -31\text{mV}$$

- Does this cause nonlinearity?

Charge Split Ratio Data



[G. Wegmann et al., IEEE JSSC, 1987]

- In practice, $R_{on} C_2$ and T_f are usually comparable
- Charge split depends on the impedances on both sides
 - More charge will go to the side with lower impedance

Summary

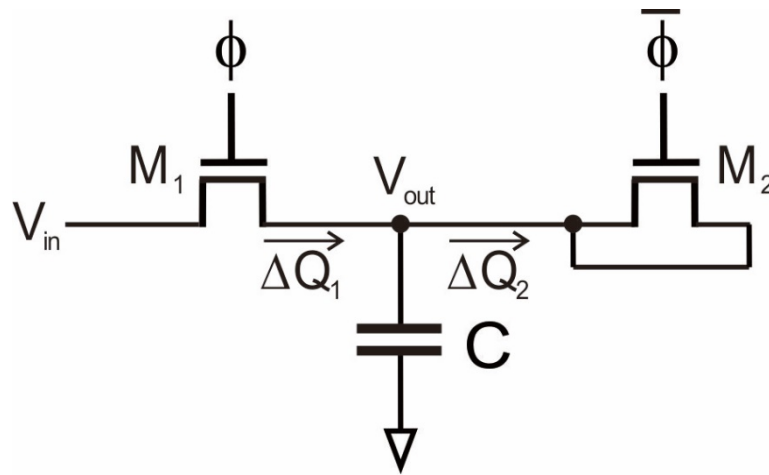
- In practice, the situation is in between slow gating and fast gating.
- Slow gating shows less gain and offset error
 - Unsuitable for high-speed
 - More later
- Are the gain error and the offset error really constant?
 - No, our first-order model is not accurate enough.
 - We have ignored back-gate effect and short-channel effect
 - In practice, about 20% or 30% of what we calculated are nonlinear errors

Does CMOS Scaling Help?

$$\left. \begin{aligned} \Delta V &\approx \frac{1}{2} \frac{Q_{ch}}{C} \\ R &\approx \frac{1}{\mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})} = \frac{L^2}{\mu Q_{ch}} \end{aligned} \right\} \Delta V \cdot R \approx \frac{L^2}{2\mu C}$$

- For the same R (speed), channel-charge induced error ΔV decreases as technology advances
 - Smaller L and larger μ
- For the same ΔV , R is smaller, leading to faster speed.
- In general, switches benefit from CMOS scaling!

Charge Cancellation



$$\Delta Q_1 \approx \frac{1}{2} Q_{ch1} + Q_{ov1}$$

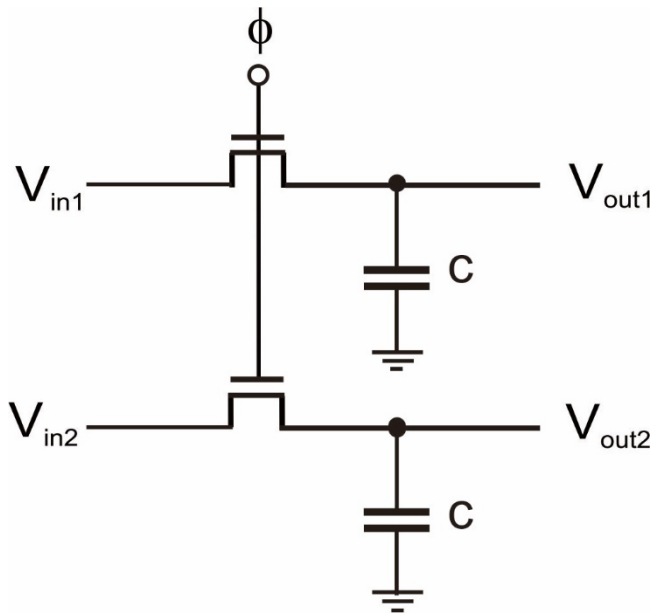
$$\Delta Q_2 \approx Q_{ch2} + 2Q_{ov2}$$

$$\Rightarrow W_2 = W_1/2, \quad L_2 = L_1$$

[Eichenberger and Guggenbühl, JSSC 8/89]

- Perfect cancellation relies on
 - Exactly 50% channel charge splitting
- Limited improvement

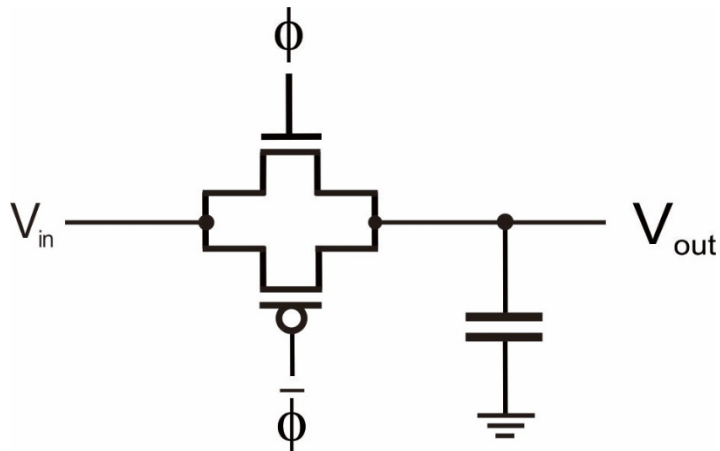
Differential Sampling



$$\begin{cases} V_{out1} = (1 + \varepsilon)V_{in1} + V_{OS} \\ V_{out2} = (1 + \varepsilon)V_{in2} + V_{OS} \end{cases}$$
$$\Rightarrow V_{out} = (1 + \varepsilon)V_{in}$$

- Offset will be cancelled, but the gain error remains the same.
 - Differential circuit only cancels out the common mode.

CMOS Switch



$$Q_{\text{chn}} \approx -W_n L_n C_{\text{ox}} (V_{\text{DD}} - V_{\text{in}} - V_{\text{thn}})$$

$$Q_{\text{chp}} \approx W_p L_p C_{\text{ox}} (V_{\text{in}} - |V_{\text{thp}}|)$$

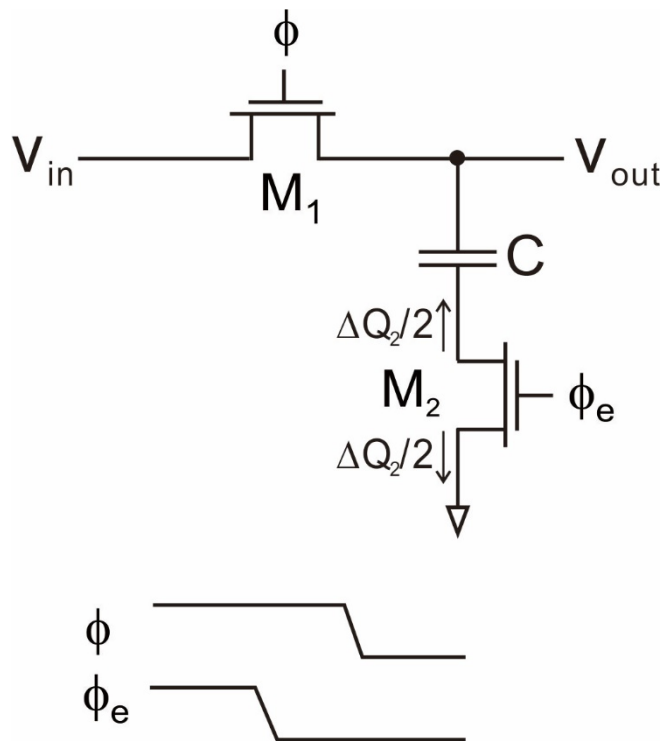
$$\Delta V_0 \approx \frac{Q_{\text{chn}} + Q_{\text{chp}}}{2C}$$

- Unfortunately, the signal dependent parts do not cancel, but add!

Bottom Plate Sampling

- Key idea: sample V_{in} at the "grounded" side of the capacitor to achieve signal independent sampling
- This technique allows us to achieve high linearity in combination with bootstrap technique
- Original papers:
 - D. J. Allstot and W. C. Black, "Technological Design Considerations for Monolithic MOS Switched-Capacitor Filtering Systems," Proc. IEEE, pp. 967-986, Aug. 1983.
 - K.-L. Lee and R. G. Meyer, "Low-Distortion Switched-Capacitor Filter Design Techniques," IEEE J. Solid-State Circuits, pp. 1103-1113, Dec. 1985.

Bottom Plate Sampling Analysis (1)



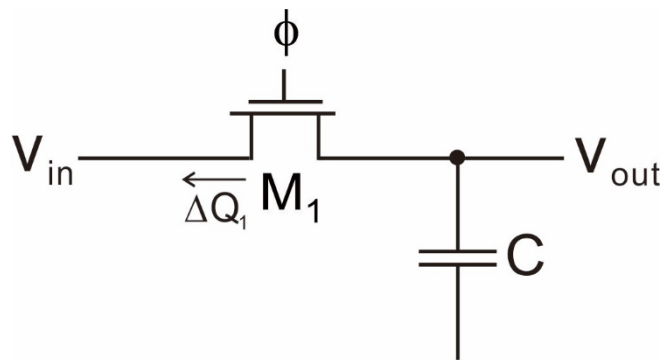
- Turn M_2 off "slightly" before M_1
 - Typically 100 ps difference
- When M_2 is off, it injects charge

$$\Delta Q_2 \approx -WLC_{ox}(V_{DD} - V_{th})$$

- To 1st-order, the charge injected by M_2 is signal independent
 - Assuming R_1 does not depend on V_{in}
- Voltage across C (50% split)

$$V_C = V_{in} - \frac{\Delta Q_2}{2C}$$

Bottom Plate Sampling Analysis (2)

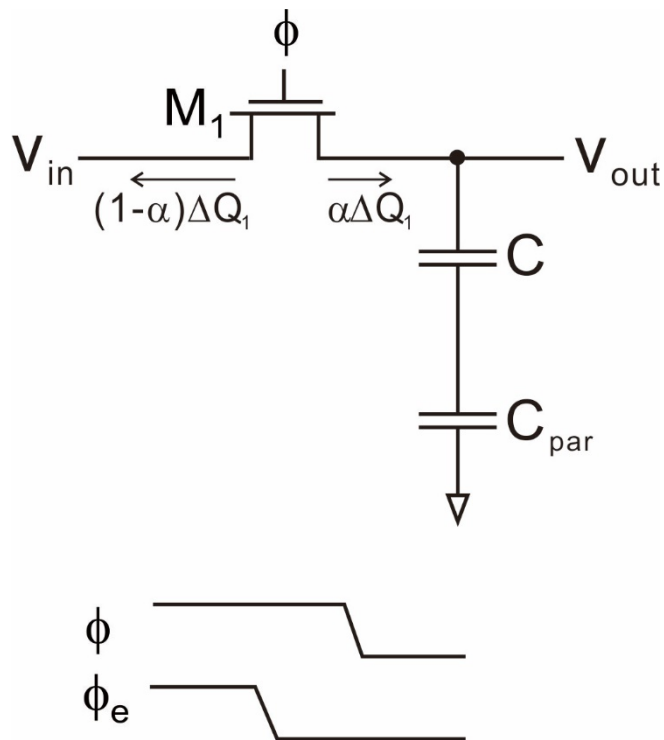


- The channel charge of M_1 is signal dependent:

$$\Delta Q_1 \approx -WLC_{ox}(\phi - V_{in} - V_{th})$$

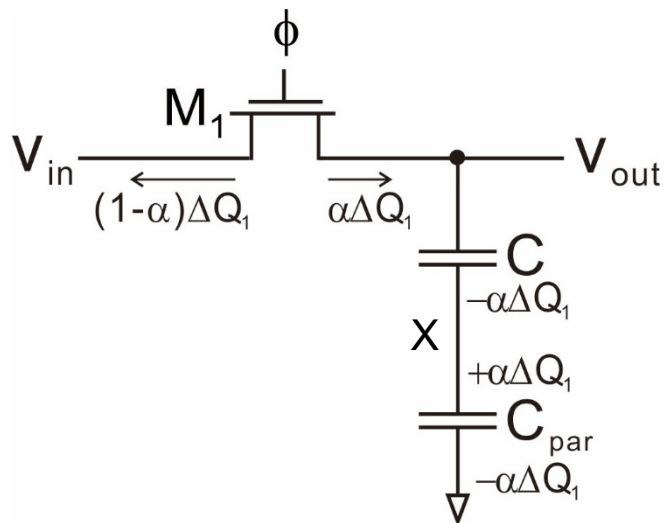
- However, when M_1 is off, it cannot inject charge to C because its bottom side is open.
- This model is too ideal – we need to consider the parasitic cap





- There are parasitic capacitors
- M1 will inject signal dependent charge onto the series combination of C and the parasitic capacitance at its bottom plate C_{par}
- This seems that the improvement is quite limited, as V_{out} is affected.
- Nevertheless, if we consider the sampled charge instead of the voltage...

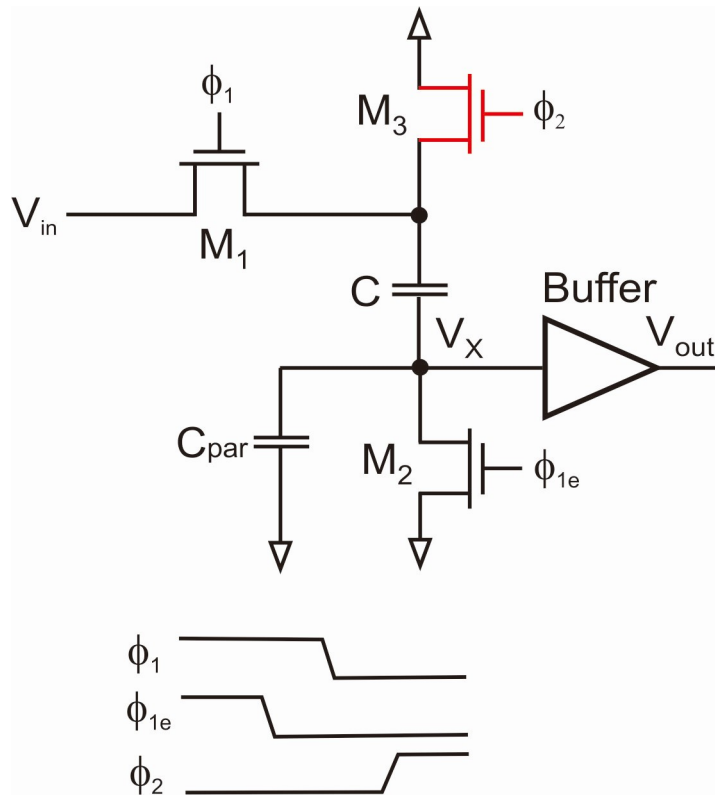
Bottom Plate Sampling Analysis (4)



- Key observation:
 - Even though M_1 injects charge to C , the total charge at node X cannot change!
- Let us process total **charge** at node X instead of looking at voltage across C .

$$Q_X = -CV_{in} + \Delta Q_2/2$$
- The charge can be processed in two ways
 - Open-loop
 - Closed-loop

Open-Loop Charge Processing



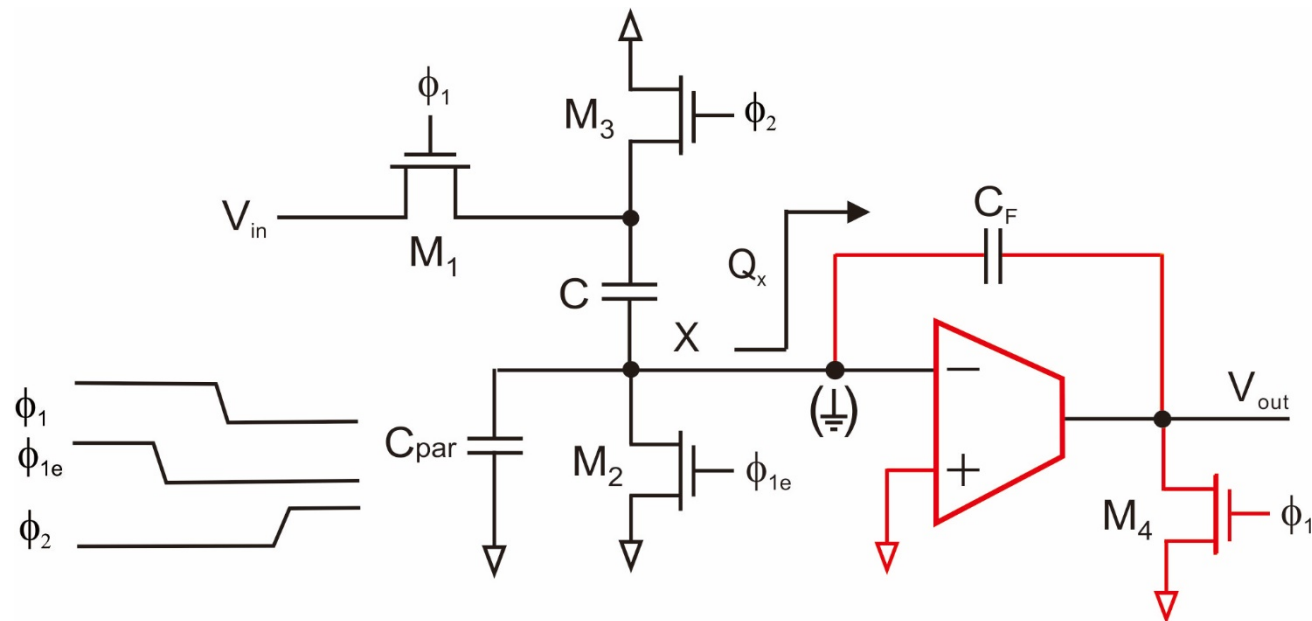
- During ϕ_1 , V_{in} is sampled at X. Q_x is linearly proportional to V_{in} .
- Due to charge conservation, Q_x does not change when M_3 is on $\rightarrow$

$$Q_x = -CV_{in} + \Delta Q_2/2 = (C + C_{par})V_x$$

$$V_{out} = V_x = -\frac{C}{C + C_{par}}V_{in} + \frac{\Delta Q_2/2}{C + C_{par}}$$

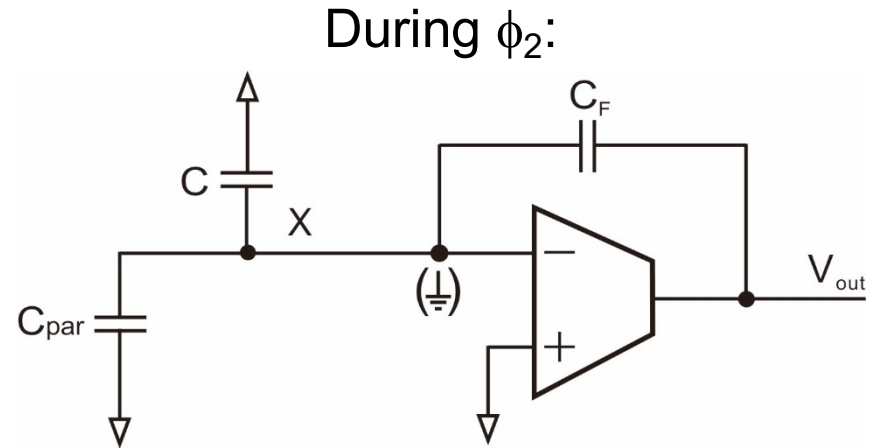
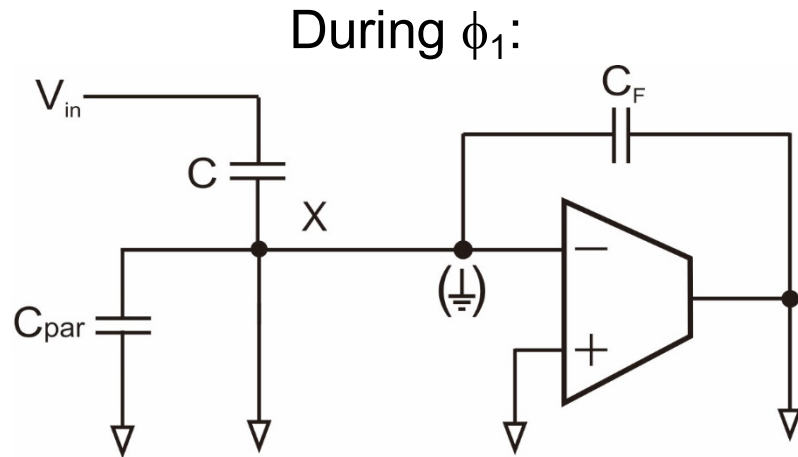
- Signal flow: $V_{in} \rightarrow Q_x \rightarrow V_x$
- Remaining drawback
 - C_{par} includes transistor parasitic cap. It is nonlinear and will introduce distortion.

Closed-Loop Charge Processing



- The opamp forces voltage at node X to “zero”
 - Means that charge at node X must redistribute onto feedback capacitor C_F

Charge Conservation Analysis



Charge at node X during when M_2 is off:

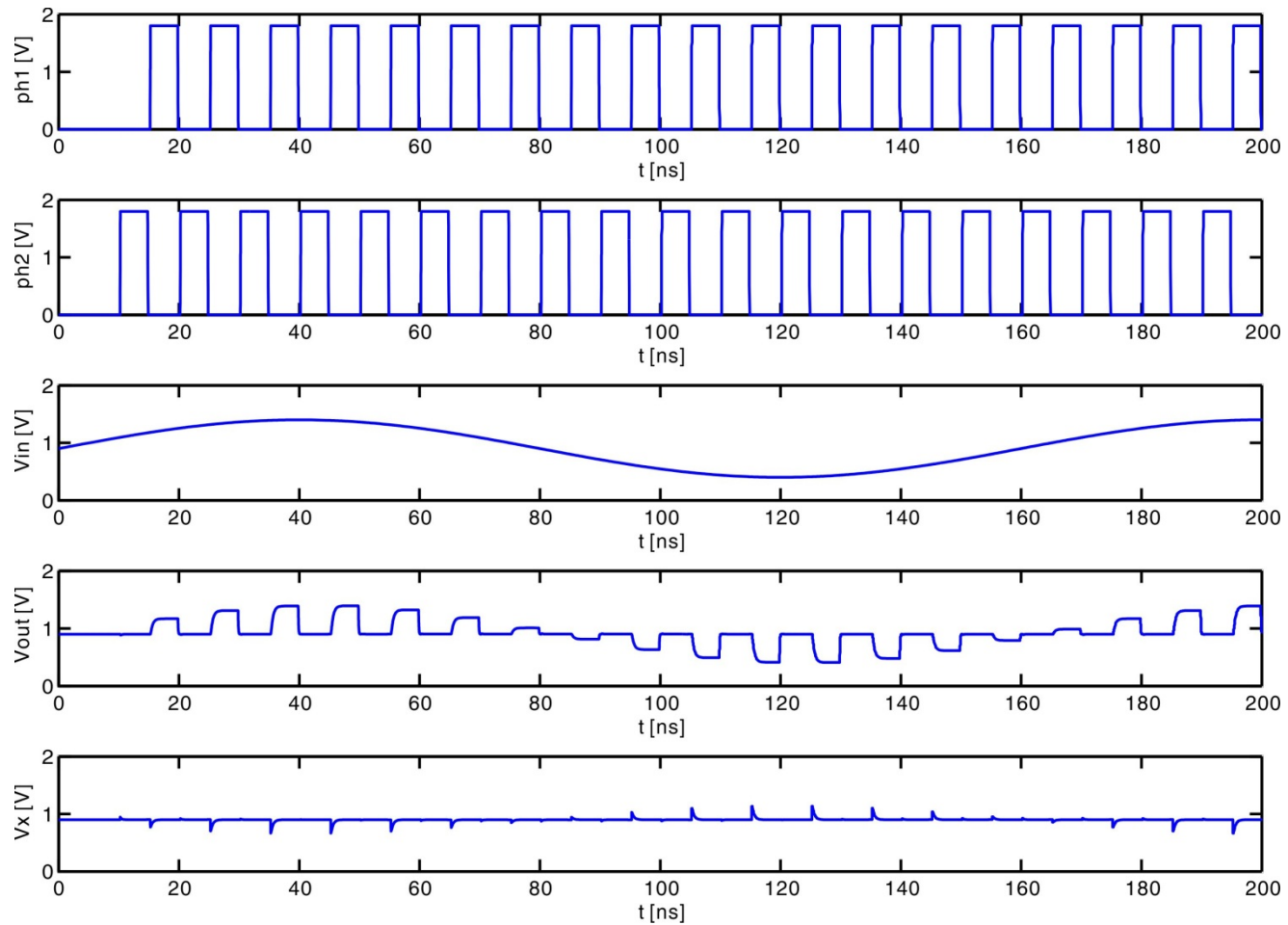
$$Q_{x1} = -CV_{in} + \Delta Q_2/2$$

Charge at node X during ϕ_2 :

$$Q_{x2} = -C_F V_{out}$$

$$Q_{x1} = Q_{x2} \Rightarrow V_{out} = \frac{C}{C_F} V_{in} + \frac{-\Delta Q_2}{2C_F}$$

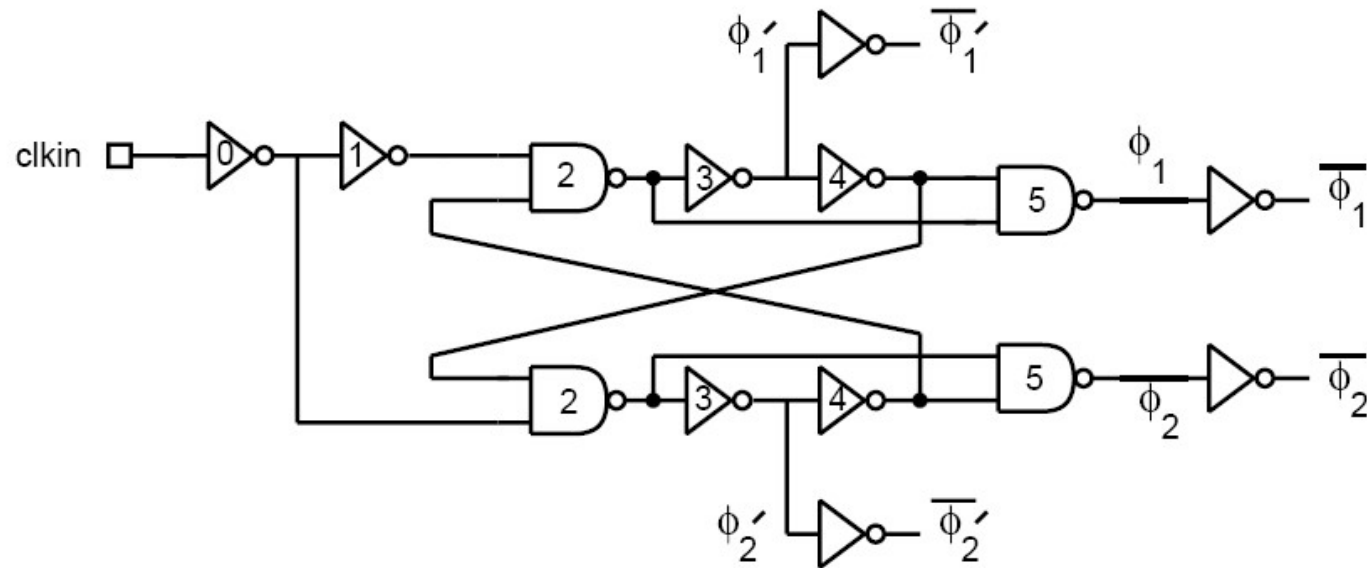
Behavioral Simulation



Comments

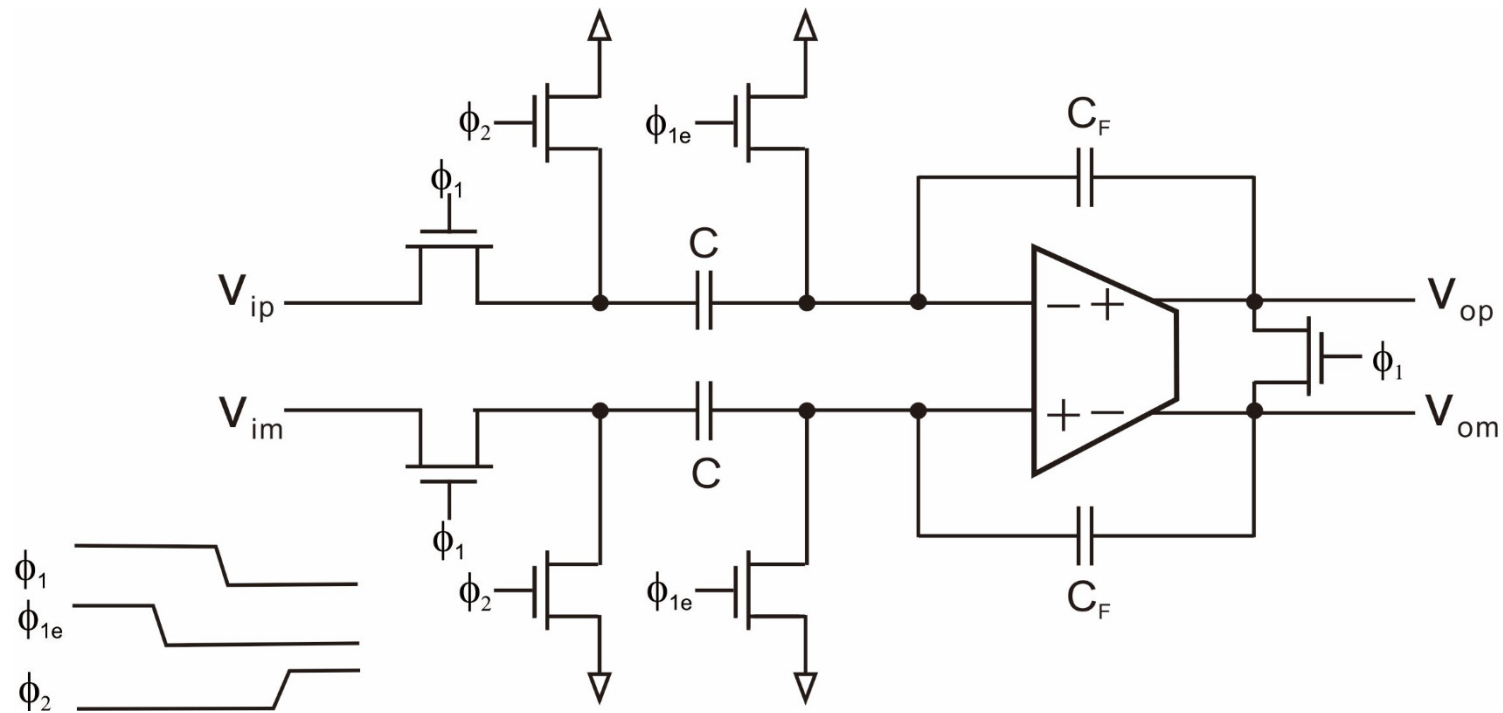
- C_F is highly linear (more later)
- Gain can be provided.
- Offset term can be easily removed by going differential.
- Linearity can be limited by opamp (offset, insufficient gain, nonlinear gain)

Clock Generation

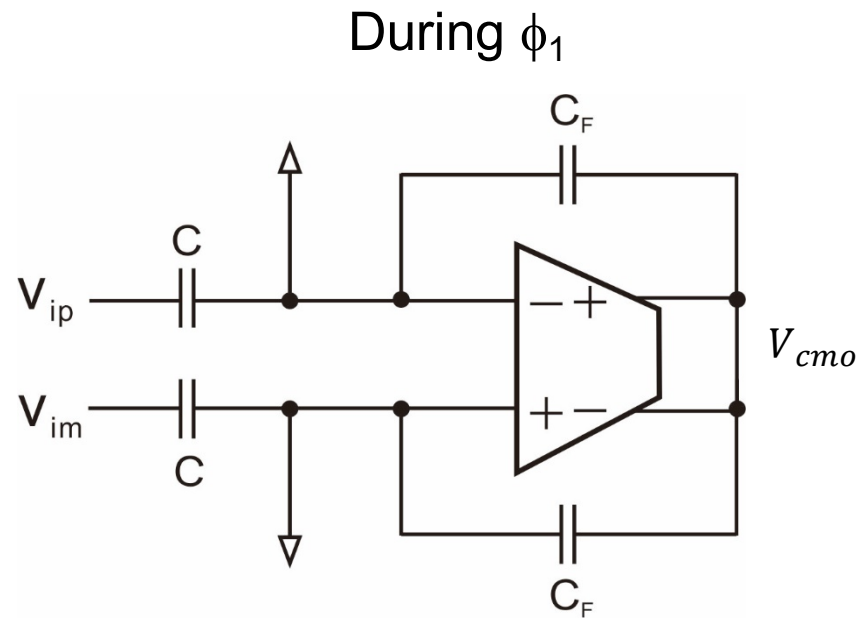


[A. Abo, "Design for Reliability of Low-voltage, Switched-capacitor Circuits," PhD Thesis, UC Berkeley, 1999]

Fully Differential Circuit

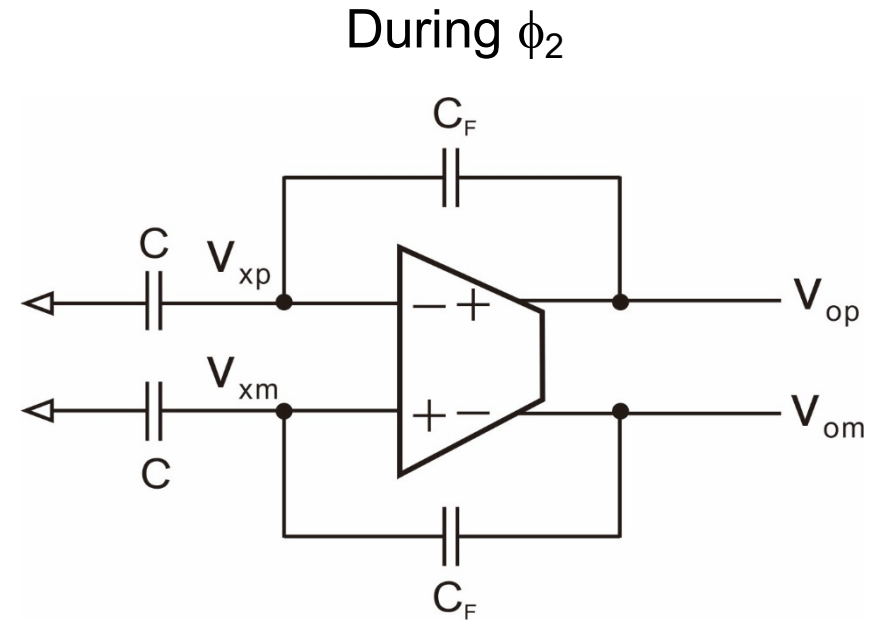


Analysis (1)



$$Q_{1m} = -CV_{im} - C_F V_{cm0} + \Delta Q$$

$$Q_{1p} = -CV_{ip} - C_F V_{cm0} + \Delta Q$$



$$Q_{2m} = CV_{xm} + C_F(V_{xm} - V_{om}) = Q_{1m} \quad (1)$$

$$\longleftrightarrow Q_{2p} = CV_{xp} + C_F(V_{xp} - V_{op}) = Q_{1p} \quad (2)$$

$$V_{xm} = V_{xp}$$

Analysis (2)

- Subtracting 1) and 2), we obtain:

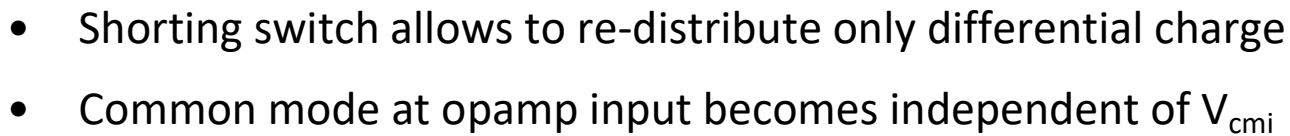
$$V_{out} = V_{op} - V_{om} = \frac{C}{C_F} (V_{ip} - V_{im})$$

- Adding 1) and 2), we obtain:

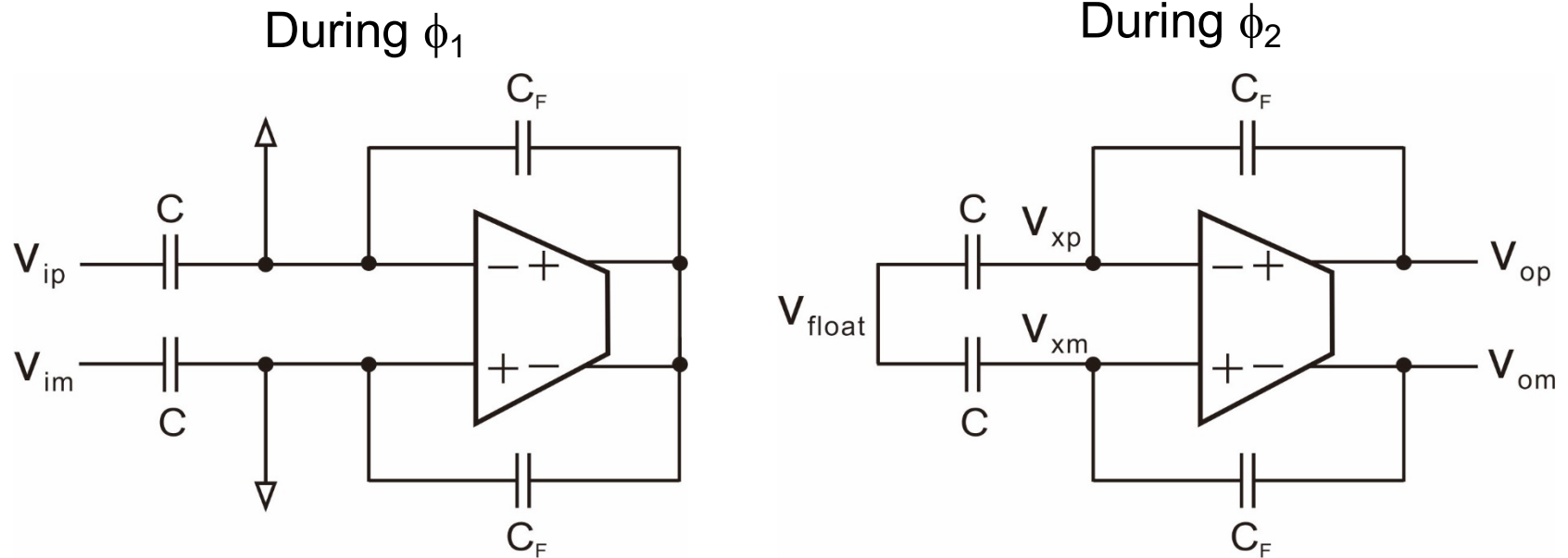
$$\begin{aligned} & -C(V_{ip} + V_{im}) - 2C_F V_{cmo} + 2\Delta Q \\ & = (C + C_F)(V_{xp} + V_{xm}) - C_F(V_{op} + V_{om}) \\ & \quad - C V_{cmi} + \Delta Q = (C + C_F)V_{cmx} \end{aligned}$$

$$\Rightarrow V_{cmx} = -\frac{C}{C + C_F} V_{cmi} + \frac{\Delta Q}{C + C_F}$$

- V_{cmi} (input common mode) shows up at the opamp input
 - Opamp must have wide input common mode range
 - Opamp must have good CMRR



Analysis (1)



- Charge conservation at the left plate of C :

$$\begin{aligned} (V_{ip} + V_{im})C &= (V_{float} - V_{xp})C + (V_{float} - V_{xm})C \\ V_{cmi} &= V_{float} - V_{cmx} \end{aligned}$$

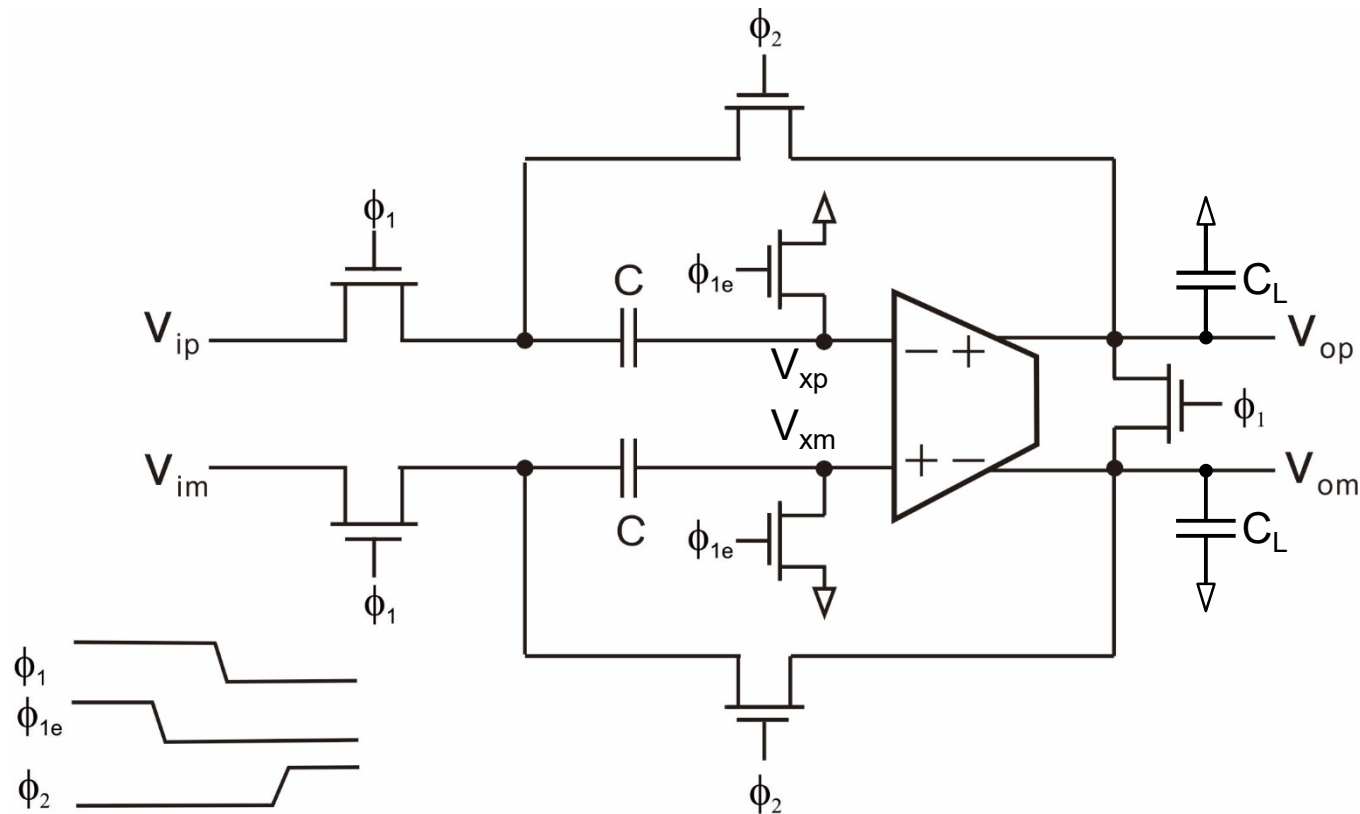
Analysis (2)

- Common mode charge conservation at the opamp input node:

$$\begin{aligned} -V_{cmi}C - V_{cmo}C_F &= -(V_{float} - V_{cmx})C - (V_{cmo} - V_{cmx})C_F \\ -V_{cmi}C &= -[(V_{cmi} + V_{cmx}) - V_{cmx}]C + V_{cmx}C_F \\ 0 &= V_{cmx} \end{aligned}$$

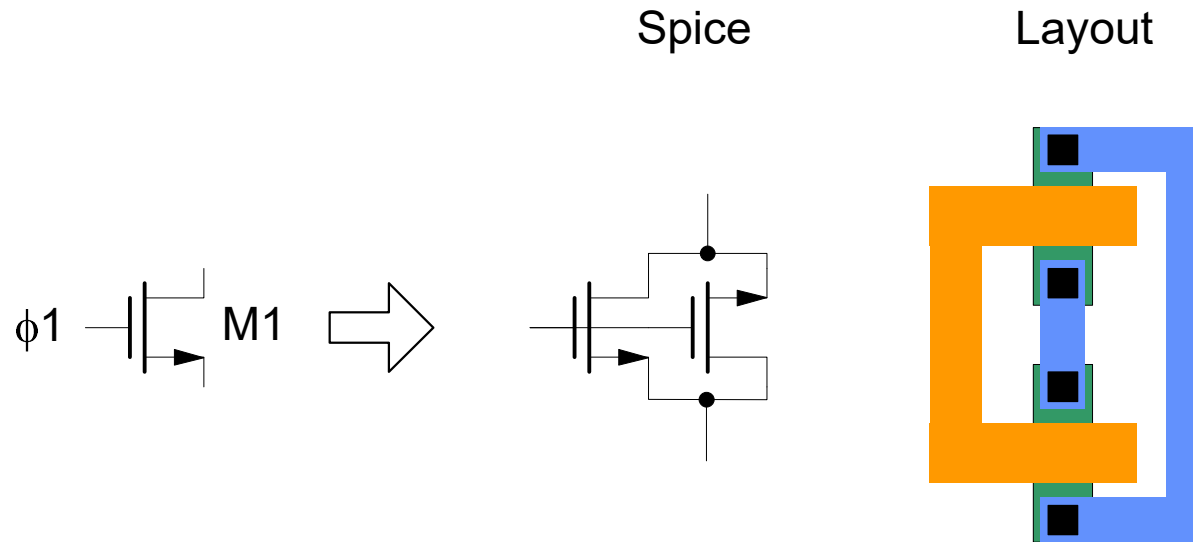
- Thus, opamp input common mode V_{cmx} is independent of
 - Input common mode V_{cmi}
 - Output common mode V_{cmo}
 - Usually set by common mode feedback circuit

Flip-Around T/H



- The same cap used for both sampling and opamp feedback
 - Larger feedback factor (lower noise, higher speed) 😊
 - OTA is subjected to input common mode variations ☹

Schematic and Layout of M_1



From B. Murmann, EE 315, Ch. 5

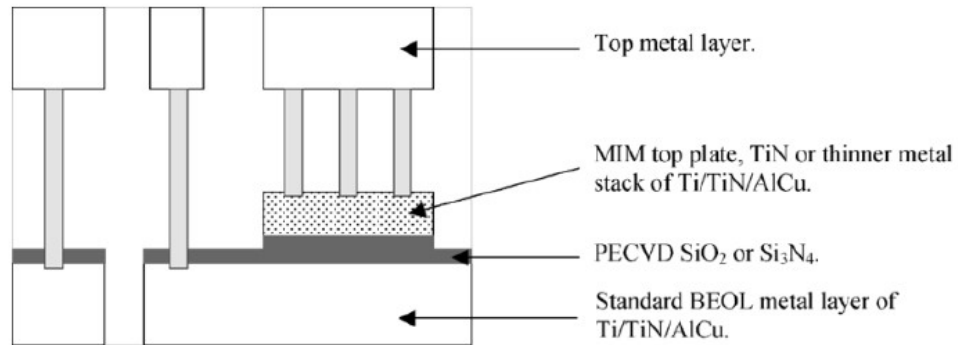
- M_1 is usually implemented using anti-parallel devices to ensure symmetry
 - Drain and source are not necessarily symmetric in circuit models
 - Drain and source are not symmetric due to process artifacts

What Ultimately Limits Linearity?

- Bottom-plate sampling plus clock bootstrapping solve majority nonlinearity problems. We can achieve 80~100dB linearity up to a few hundreds of MHz.
- Limitation of clock bootstrapping: $R_{top,SW}$ varies with V_{in} due to **back-gate effects** and **parasitic capacitances** (especially at high frequency)
 - In the track phase, it causes **nonlinear filter response**
 - Bottom-plate sampling does not help.
 - In the hold phase, it causes **input dependent charge injection** because the charge splitting depends on the impedance
 - We can mitigate this issue by reducing $R_{top,SW}$
- Limitation of bottom-plate sampling
 - Open-loop charge transfer: nonlinear parasitic cap
 - Close-loop charge transfer: opamp nonlinearity

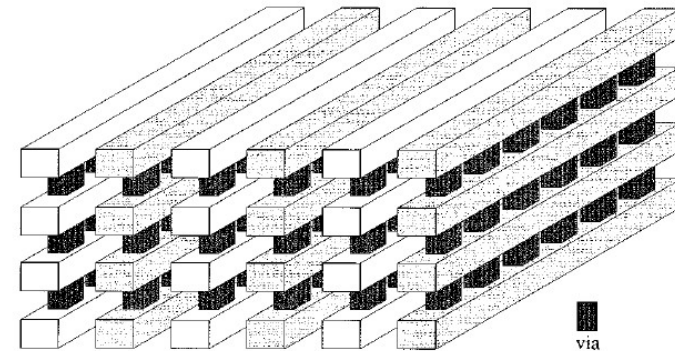
Capacitors

Metal-Insulator-Metal (MIM)



[Ng, Trans. Electron Dev., 7/2005]

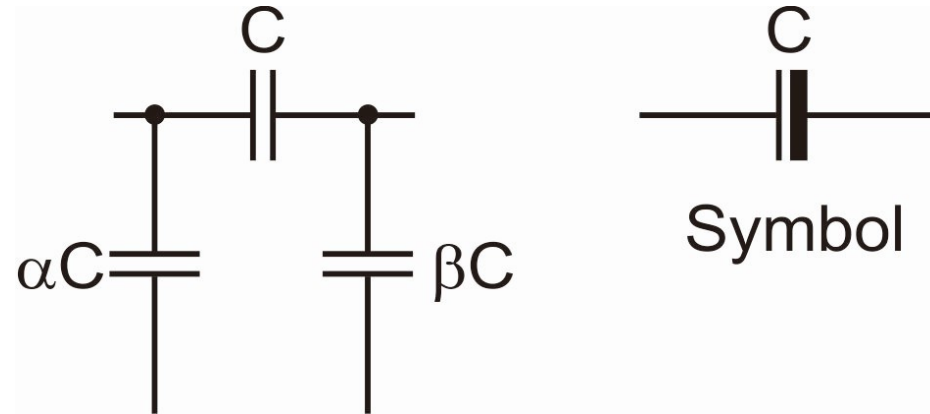
Metal-On-Metal (MOM)



[Aparicio, JSSC 3/2002]

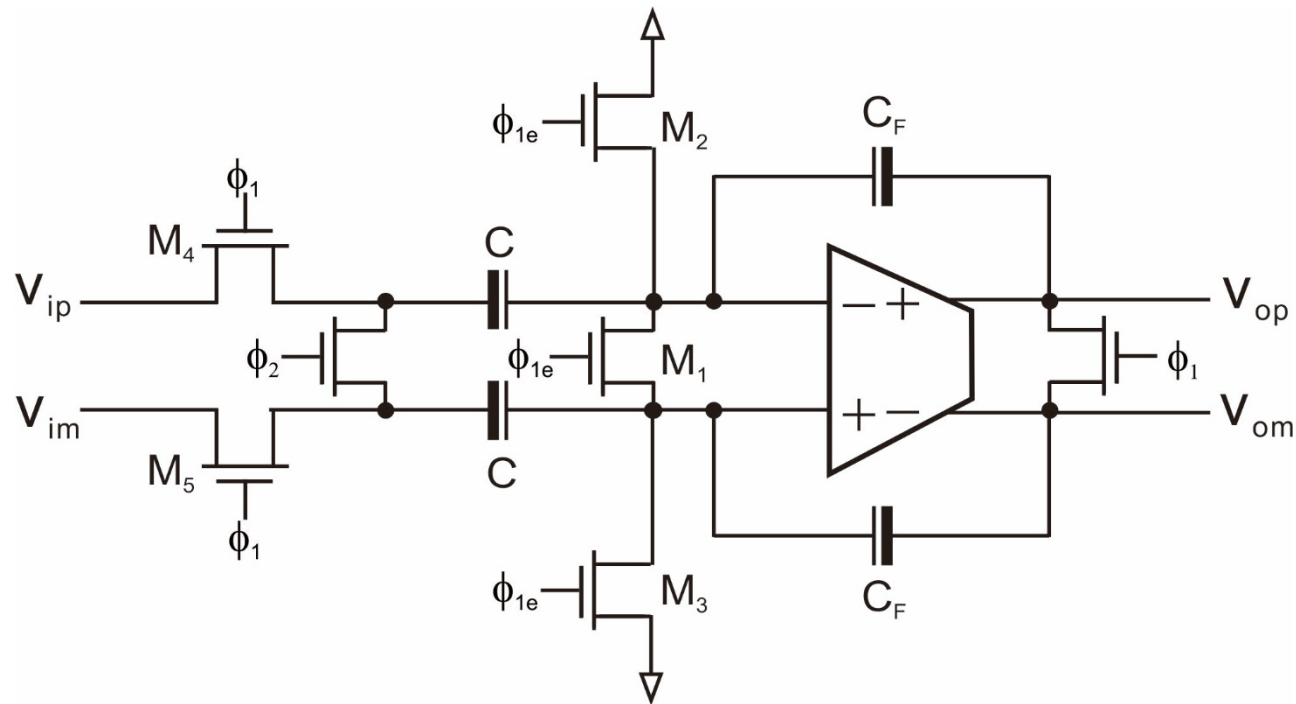
- Both MIM and MOM capacitors have good electrical properties
 - Series and parallel resistances are usually not a concern
 - Mostly worry about parasitic caps
- MOM caps are preferred in advanced processes due to higher density.
 - The density increases as process scales.
- MIM cap density is usually 1-2 fF/ μm^2
 - For 1 fF/ μm^2 , a 10 pF capacitor occupies 100 μm x 100 μm

Plate Parasitics



- The two plates of a real capacitor are usually unsymmetrical
 - We usually use the fat plate to indicate the plate that has larger parasitic capacitance.
- For a MIM capacitor, the plate that is away from the substrate has smaller parasitic capacitance (typical value: $\alpha=1\%$, $\beta=10\%$).
- For a MOM capacitor, α and β can vary.
- We can change the effective α and β through layout design.

Proper Connection of Capacitors



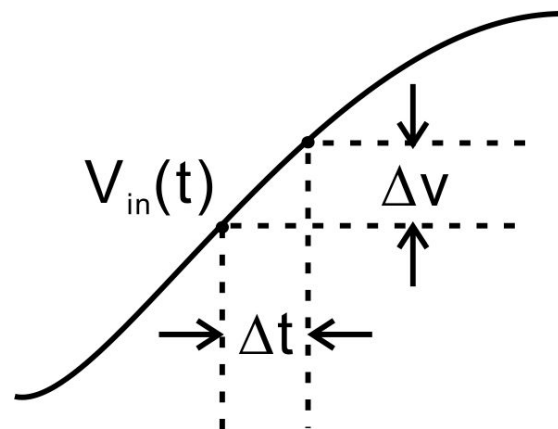
- The more ideal plate should face the virtual ground nodes
 - Avoid reduction of feedback factor
 - Reduce potential noise coupling

Nonidealities

- Switch resistance
- Switch capacitance
- Sampling clock jitter
- Other nonidealities

Sampling Clock Jitter

- Clock jitter introduces sampling errors.
 - Its magnitude is proportional to the variation of V_{in} .



$$\Delta V_{in} \approx \frac{dV_{in}}{dt} \cdot \Delta t$$

- Let us consider a sine wave input signal and assume Δt follows normal distribution with zero mean and standard deviation σ_t

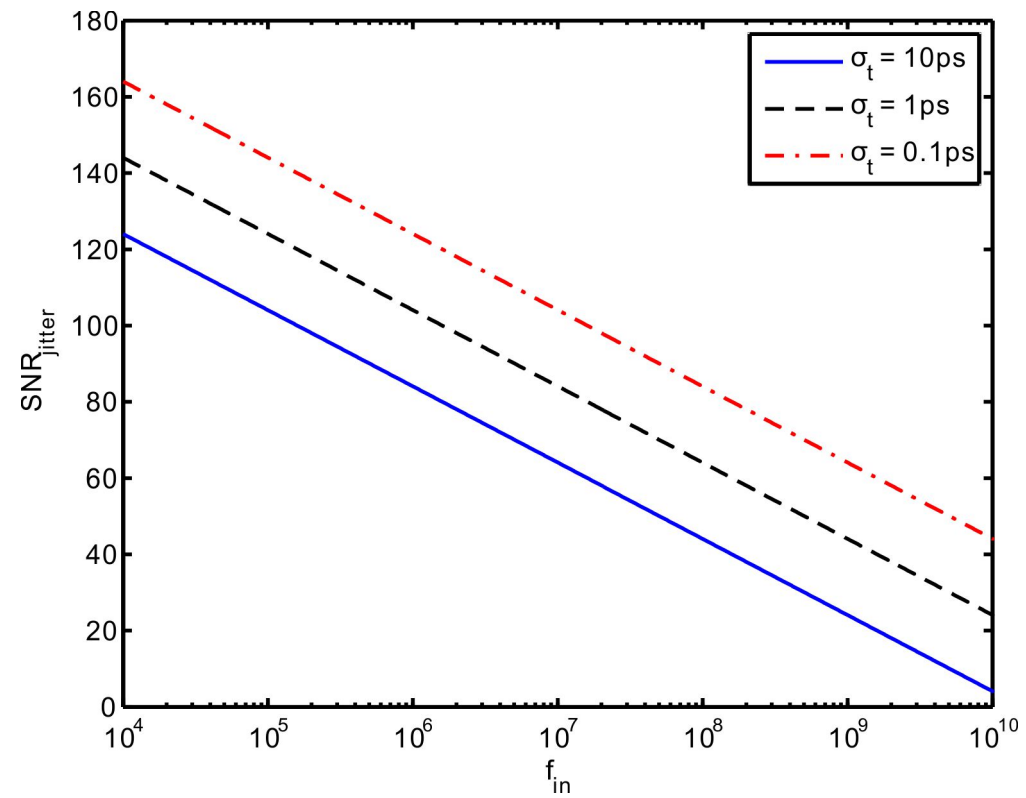
Analysis

$$\begin{aligned} E\{\Delta V_{in}^2\} &\cong E\left\{\left(\frac{dV_{in}}{dt}\right)^2 \cdot \Delta t^2\right\} = E\left\{\left(\frac{dV_{in}}{dt}\right)^2\right\} \cdot E\{\Delta t^2\} \\ &\cong E\left\{\left(\frac{d}{dt} A \cos[2\pi \cdot f_{in} \cdot t]\right)^2\right\} \cdot \sigma_t^2 \cong \frac{1}{2} (2\pi \cdot A \cdot f_{in})^2 \cdot \sigma_t^2 \end{aligned}$$

$$SNR_{jitter} [dB] \cong 10 \cdot \log \left[\frac{\frac{1}{2} A^2}{\frac{1}{2} (2\pi \cdot A \cdot f_{in} \cdot \sigma_t)^2} \right] = 20 \cdot \log \left[\frac{1}{2\pi \cdot f_{in} \cdot \sigma_t} \right]$$

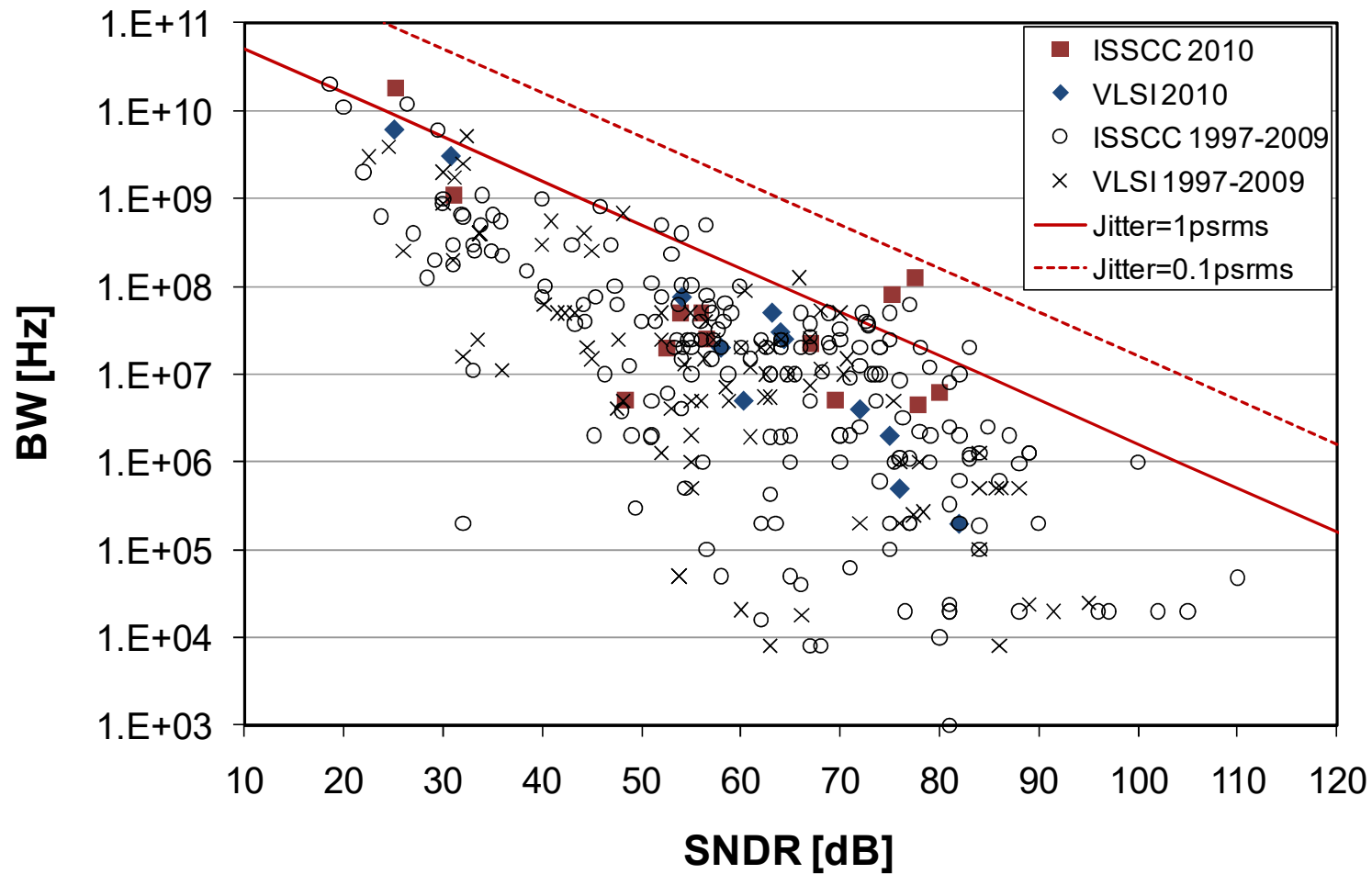
- Note that this jitter limited SNR does not depend on signal amplitude!
- The jitter induced error increases as signal amplitude increases.
 - In testing, if SNR remains constant for various signal amplitudes, it is highly possible that the SNR is limited by clock jitter.

Result



ADC Performance Survey

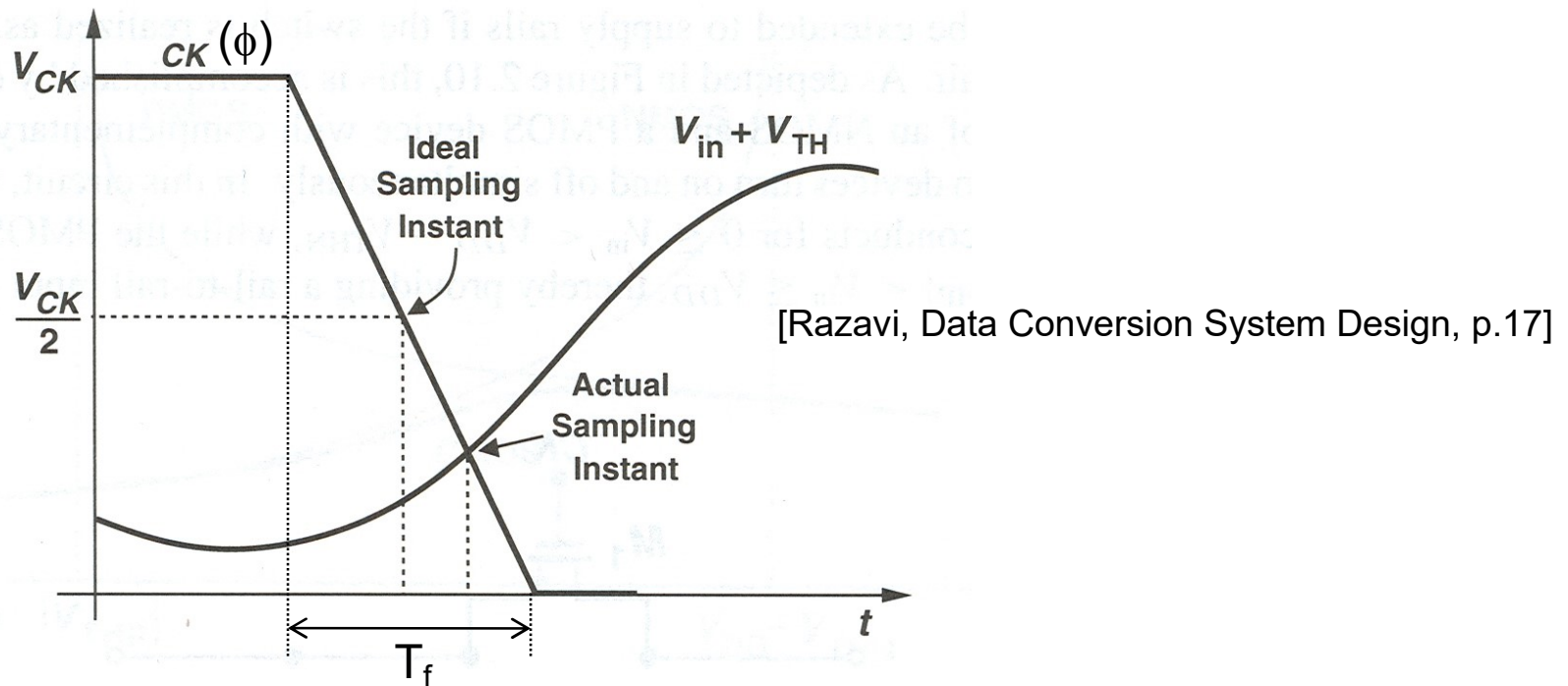
From B. Murmann, EE 315, Ch. 5



Nonidealities

- Switch resistance
- Switch capacitance
- Sampling clock jitter
- Other nonidealities
 - Signal dependent sampling instants
 - Hold mode feed-through
 - Hold mode leakage

Signal Dependent Sampling Instant (1)



- Must make T_f much smaller than maximum dV_{in}/dt
 - This means that we are in the fast-gating scenario.

Signal Dependent Sampling Instant (2)

- Distortion analysis result (see Yu, TCAS II, 2/1999]

$$HD_3 = \frac{3}{8} \left(\frac{A}{V_{CLK}} \omega T_F \right)^2$$

- Example #1: $V_{CLK} = 1.8V$, $A = 0.5V$, $T_f = 100ps$, $\omega = 2\pi \times 100MHz$

$$HD_3 = \frac{3}{8} \left(\frac{0.5}{1.8} 2\pi \cdot 100e6 \cdot 100e-12 \right)^2 = -79dB$$

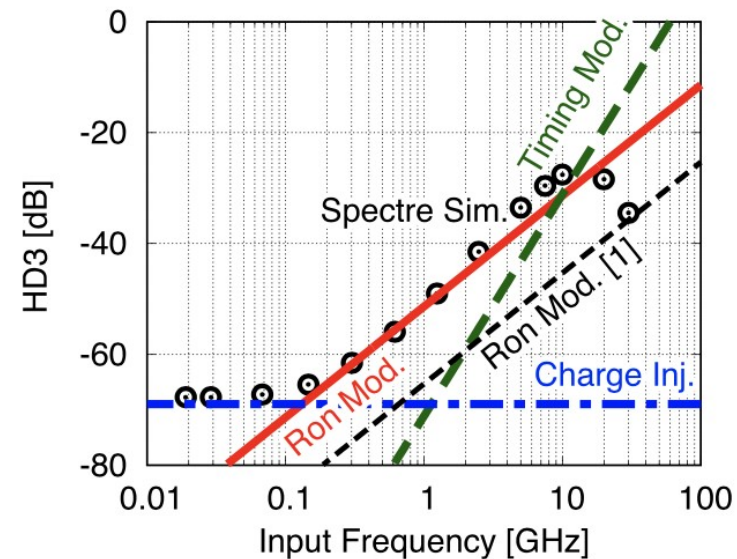
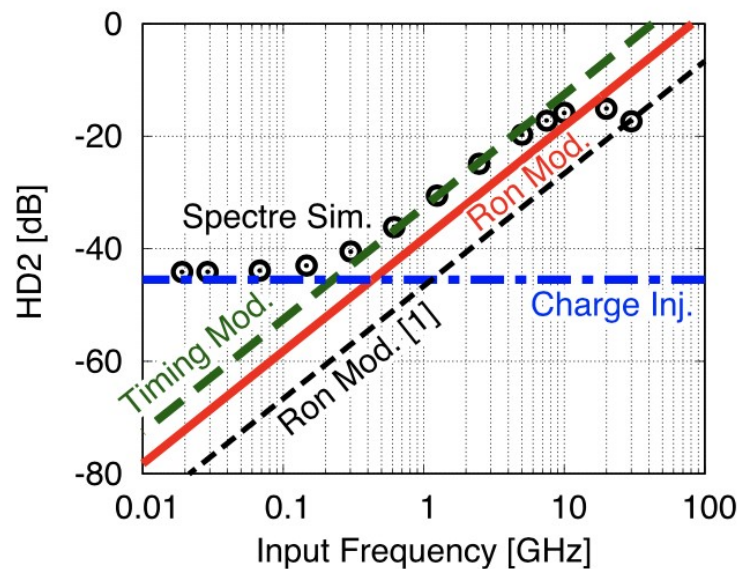
- Example #2: $V_{CLK} = 1.8V$, $A = 0.5V$, $T_f = 100ps$, $\omega = 2\pi \times 1GHz$

$$HD_3 = \frac{3}{8} \left(\frac{0.5}{1.8} 2\pi \cdot 1e9 \cdot 100e-12 \right)^2 = -39dB$$

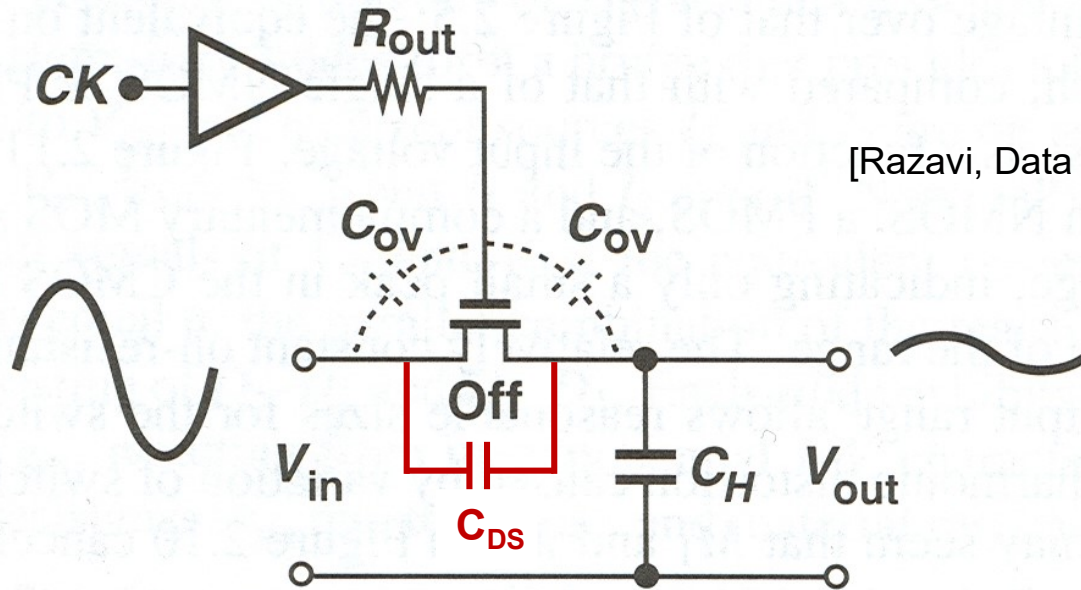
- Short T_f is a must for sampling high frequency signal
 - Note that a realistic T_f should be set in simulation

Signal Dependent Sampling Instant (3)

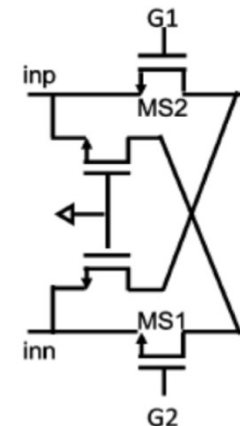
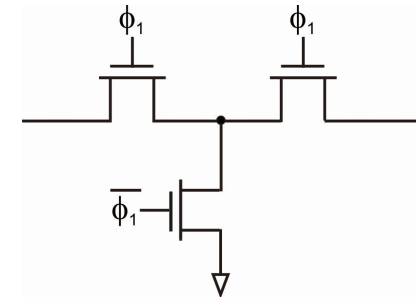
- Distortion due to signal dependent sampling instant (also called timing modulation, green line in the figures below), dominates at high input frequency compared to charge-injection-based distortion and R_{on} -modulation effect
 - T. Iizuka, T. Ito and A. A. Abidi, "Comprehensive Analysis of Distortion in the Passive FET Sample-and-Hold Circuit"



Hold Mode Feedthrough



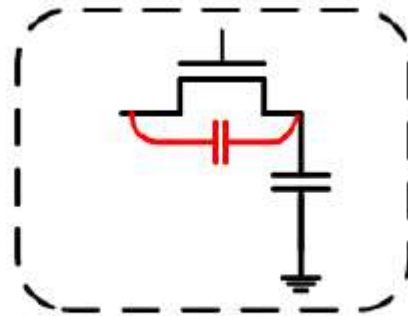
[Razavi, Data Conversion System Design, p.17]



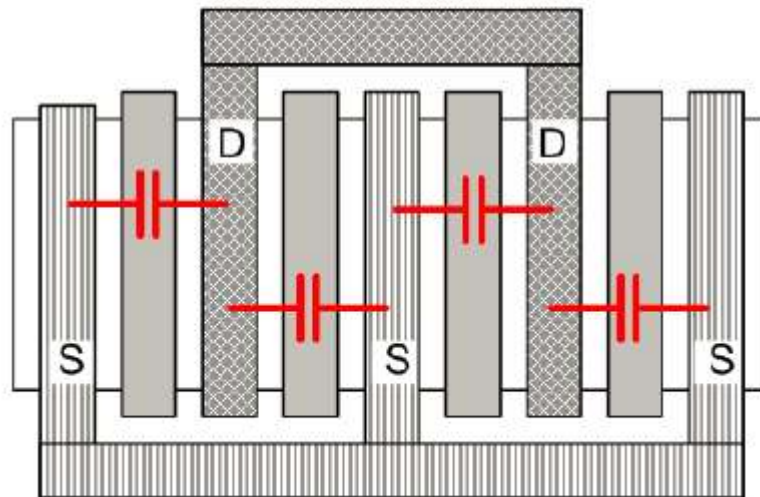
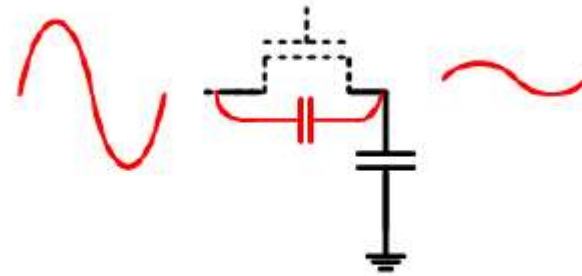
- Make R_{out} as small as possible
- Use T-switch when hold-mode feedthrough is a problem (for low-speed)
- Use cross coupling for cancellation (for high-speed)
- Minimize C_{DS} in layout

Minimize C_{DS} (1)

NMOS sampling switch



[Manar El-Chammas, ISSCC 2015, 15.8 slide]

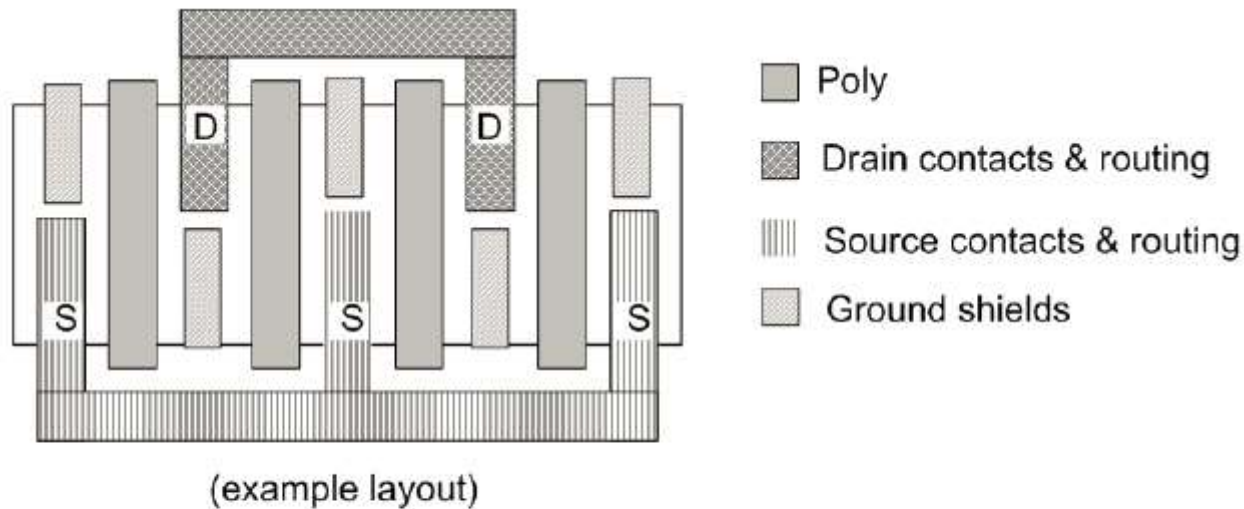


(example layout)

- Poly (routing not shown)
- Drain contacts & routing
- Source contacts & routing

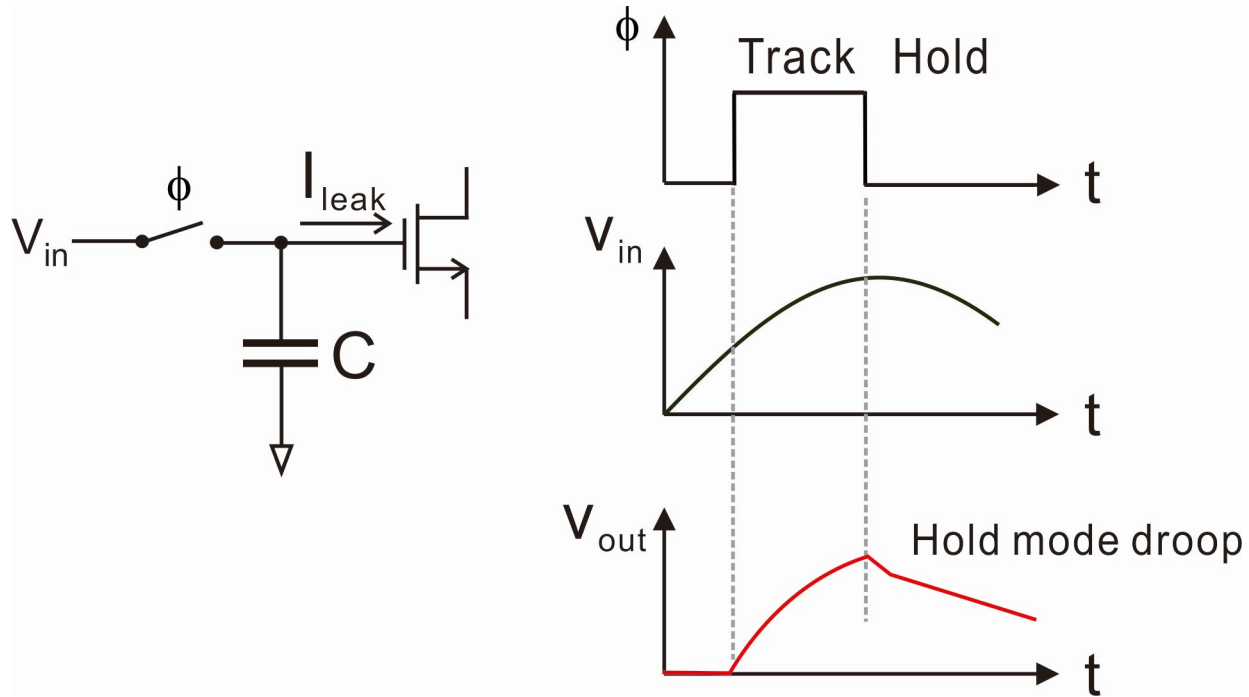
Minimize C_{DS} (2)

[Manar El-Chammas, ISSCC 2015, 15.8 slide]

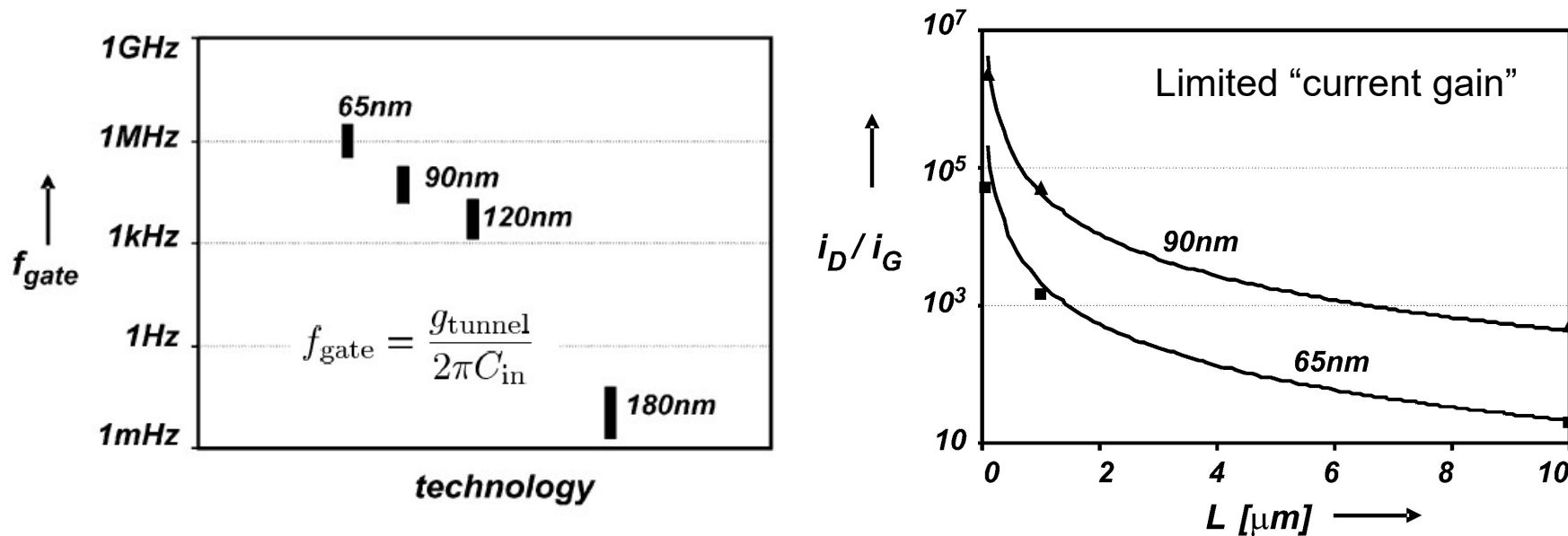


- Reduce contacts $\rightarrow$ capacitance cut $\sim 15X$
- Add ground shields $\rightarrow > 30X$ reduction

Hold Mode Leakage



Gate Leakage Data



A. Annema, et al., "Analog circuits in ultra-deep-submicron CMOS," *IEEE J. Solid-State Circuits*, pp. 132-143, Jan. 2005.

- f_{gate} indicates how fast charge leaks
- Gate leakage depends strongly on dielectric thickness (tunneling effect)
 - 1nm reduction in t_{ox} leads to 10^6 increase in gate leakage current
- The use of high-K dielectric (H_f) reduces gate leakage
- Gate leakage is more troublesome for low speed operation!
 - Avoid holding charges for a long time